

Supplementary Online Content

Writing Committee for the PROBESE Collaborative Group of the PROtective VEntilation Network (PROVENet) for the Clinical Trial Network of the European Society of Anaesthesiology. Effect of intraoperative high positive end-expiratory pressure (PEEP) with recruitment maneuvers vs low PEEP on postoperative pulmonary complications in obese patients: a randomized clinical trial. *JAMA*. doi:10.1001/jama.2019.7505

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Lists of committees, investigators and study sites

1.1 Writing committee members

The writing committee was composed by Thomas Bluth (Pulmonary Engineering Group, Department of Anesthesiology and Intensive Care, University Hospital Carl Gustav Carus, Technische Universität Dresden, Dresden, Germany), Ary Serpa Neto (Department of Critical Care Medicine, Hospital Israelita Albert Einstein, Sao Paulo, Brazil), Marcus J. Schultz (Department of Intensive Care, Academical Medical Center, University of Amsterdam, The Netherlands), Paolo Pelosi (Department of Surgical Sciences and Integrated Diagnostics, University of Genoa, Policlinico San Martino, Genoa, Italy) and Marcelo Gama de Abreu (Pulmonary Engineering Group, Department of Anesthesiology and Intensive Care, University Hospital Carl Gustav Carus, Technische Universität Dresden, Dresden, Germany). All members of the writing committee contributed equally.

1.2 Steering committee members

(names are listed in alphabetical order)

Bluth, Thomas	Department of Anesthesiology and Intensive Care, Pulmonary Engineering Group, University Hospital Carl Gustav Carus, Technische Universität Dresden, Germany
Bobek, Ilona	Aneszteziológiai és Intenzív Terápiás Klinika, Semmelweis Egyetem, Hungary
Canet, Jaume	University Hospital Germans Trias i Pujol, ICS, UAB, Spain
Cinnella, Gilda	Department of Anesthesiology and Intensive Care Medicine, University of Foggia, Italy
de Baerdemaeker, Luc	Ghent University Hospital, Ghent, Belgium. Corneel Heymanslaan 10, 9000 Gent, Belgium
Gama de Abreu, Marcelo	Department of Anesthesiology and Intensive Care, Pulmonary Engineering Group, University Hospital Carl Gustav Carus, Technische Universität Dresden, Germany
Gregoret, Cesare	Department of Anesthesiology, Città della Salute e della Scienza, Turin, Italy
Hedenstierna, Göran	Department of Medical Sciences, Section of Clinical Physiology, University Hospital, Uppsala, Sweden
Hemmes, Sabine	Amsterdam UMC, University of Amsterdam, Department of Anesthesiology, Meibergdreef 9, Amsterdam, The Netherlands
Hiesmayr, Michael	Division Cardiac, Thoracic, Vascular Anesthesia and Intensive Care, Medical University, Vienna, Austria
Hollmann, Markus W.	Amsterdam UMC, University of Amsterdam, Department of Anesthesiology, Meibergdreef 9, Amsterdam, The Netherlands
Jaber, Samir	Department of Critical Care Medicine and Anesthesiology (SAR B), Saint Eloi University Hospital, Montpellier, France
Laffey, John	Department of Anesthesiology, St. Michael's Hospital, University of Toronto, Toronto, ON, Canada
Licker, Marc-Joseph	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie; Faculty of Medicine, University of Geneva, Switzerland

Markstaller, Klaus	Department of Anaesthesia, Critical Care and Pain Medicine, Medical University Vienna, Austria
Matot, Idit	Division of Anesthesia, Pain and Critical Care, Tel-Aviv Medical Center affiliated with Sackler Medical School, Tel Aviv University, Tel Aviv, Israel
Mills, Gary H	Operating Services, Critical Care and Anaesthesia (OSCCA), Sheffield Teaching Hospitals and University of Sheffield, United Kingdom
Mulier, Jan Paul	Department of Anesthesiology, AZ Sint Jan Brugge-Oostende AV, Belgium
Pelosi, Paolo	Dipartimento di Scienze Chirurgiche e Diagnostiche Integrate – DISC – Università degli Studi di Genova – Genova – Italy and Anestesia e Terapia Intensiva - IRCCS Policlinico San Martino – Genova – Italy
Putensen, Christian	Department of Anaesthesiology and Intensive Care Medicine, University Hospital of Bonn, Bonn, Germany
Rossaint, Rolf	Department of Anesthesiology, Medical Faculty, RWTH Aachen University, Aachen, Germany
Schmitt, Jochen	Center for Evidence-based Healthcare, University Hospital Dresden, Germany
Schultz, Marcus J.	Amsterdam UMC, University of Amsterdam, Department of Anesthesiology, Meibergdreef 9, Amsterdam, The Netherlands
Senturk, Mert	Istanbul University, Istanbul Faculty of Medicine, Turkey
Serpa Neto, Ary	Department of Critical Care Medicine, Hospital Israelita Albert Einstein, Sao Paulo, Brazil
Severgnini, Paolo	Università della Insubria – Dipartimento di Anestesia - Azienda Ospedaliera asst Settelaghi Ospedale di Circolo e Fondazione Macchi – 21100 Varese - Italy
Sprung, Juraj	Department of Anesthesiology and Perioperative Medicine, Mayo Clinic, Rochester, Minnesota, USA
Vidal Melo, Marcos	Department of Anesthesia, Critical Care and Pain Medicine, Massachusetts General Hospital, Boston, Massachusetts, USA
Wrigge, Hermann	Department of Anaesthesiology and Intensive Care Medicine, University of Leipzig, Leipzig, Germany and Department of Anaesthesiology, Intensive Care and Emergency Medicine, Pain Therapy; Bergmannstrost Hospital Halle, Halle, Germany

1.3 Statistician

Ary Serpa Neto (Department of Critical Care Medicine, Hospital Israelita Albert Einstein, Sao Paulo, Brazil).

1.4 Data safety and monitoring committee

Patient data and safety was monitored by a committee, which was composed of a chairperson (Daniel Sessler) and four further members (Jennifer Hunter, Jeanine Wiener-Kronish, Jean-Louis Vincent and Andreas Hoeft). All adverse events (AEs) were entered into the electronic clinical report form within pre-specified time frames, including severe AEs and suspected unexpected severe adverse events, were monitored by an international AE-manager (Ary Serpa Neto), who provided the data and safety monitoring committee with reports for review. The

committee further monitored the overall status of the trial, e.g. progress of patient enrollment, general adherence to protocol and completeness of data entry.

1.5 PROBESE investigators

(*, indicates local principal investigators; names are listed in alphabetical order)

Abelha, Fernando*	Centro Hospitalar de São João, Department of Anaesthesiology, Porto, Portugal and Department of Surgery and Physiology, Cardiovascular Research Center, Faculty of Medicine of the University of Porto, Porto, Portugal
Abitağaoğlu, Sühayla	Fatih Sultan Mehmet Educational and Research Hospital, İstanbul, Turkey
Achilles, Marc*	Marienhospital Wesel, Wesel, Germany
Adebesin, Afeez	Imperial College Healthcare NHS Trust, UK
Adriaenssens, Ine	University Hospital Antwerp, Belgium
Ahene, Charles*	Cleveland Clinic Abu Dhabi, Al Maryah Island, Abu Dhabi, UAE
Akbar, Fatima	Imperial College Healthcare NHS Trust, UK
Al Harbi, Mohammed	Ministry of National Guard Health Affairs, King Abdulaziz Medical City – Riyadh, Anesthesia Department and King Saud Bin Abdulaziz University for Health science, Saudi Arabia
Al Khoury al Kallab, Rita	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Albanel, Xavier	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Aldenkortt, Florence	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Alfouzan, Rawan	Ministry of National Guard Health Affairs, King Abdulaziz Medical City – Riyadh, Anesthesia Department, Saudi Arabia
Abdullah Saleh	Ministry of National Guard Health Affairs, King Abdulaziz Medical City – Riyadh, Anesthesia Department, Saudi Arabia
Alruqaie, Reef	Ministry of National Guard Health Affairs, King Abdulaziz Medical City – Riyadh, Anesthesia Department, Saudi Arabia
Altermatt, Fernando*	Department of Anesthesiology. School of Medicine. Pontificia Universidad Catolica de Chile
Araujo, Bruno Luís de Castro*	Department of Anesthesiology, Hospital do Câncer II, National Cancer Institute of Brazil (INCA), Rio de Janeiro, Brazil
Arbesú, Genaro	University Hospital Germans Trias i Pujol, ICS, UAB, Spain
Artsi, Hanna	Division of Anesthesia, Pain and Critical Care, Tel-Aviv Medical Center affiliated with Sackler Medical School, Tel Aviv University, Tel Aviv, Israel
Aurilio, Caterina*	Department of Women, child and General and Specialized Surgery. University of Campania "L. Vanvitelli", Italy.
Ayanoglu, Omer Hilmi	Marmara University Pendik Training and Research Hospital, Turkey
Bacuzzi, Alessandro	Università dell'Insubria – Dipartimento di Anestesia - Azienda Ospedaliera Asst Settlaghi Ospedale di Circolo e Fondazione Macchi – 21100 Varese - Italy
Baig, Harris*	University of Mississippi Medical Center, MS, USA
Baird, Yolanda	St Richard's Hospital, Chichester, UK
Balonov, Konstantin*	Department of Anesthesiology and Perioperative Medicine, Tufts Medical Center, Boston, MA, USA

Balust, Jaume	Department of Anaesthesiology, Hospital Clínic, Barcelona, Spain.
Banks, Samantha	Royal Cornwall Hospital NHS Trust, UK
Bao, Xiaodong	Department of Anesthesia, Critical Care and Pain Medicine, Massachusetts General Hospital, Boston, Massachusetts, USA
Baumgartner, Mélanie	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Belda Tortosa, Isabel*	Hospital Sagrat Cor- Barcelona, Spain
Bergamaschi, Alice	Department of Anesthesiology and Intensive Care, Pulmonary Engineering Group, University Hospital Carl Gustav Carus, Technische Universität Dresden, Germany
Bergmann, Lars*	Klinik für Anästhesiologie, Intensivmedizin und Schmerztherapie, Universitätsklinikum Knappschafts Krankenhaus Bochum, Bochum, Germany
Bigatello, Luca*	St. Elizabeth's Medical Center, MA, USA
Biosca Pérez, Elena	Consorcio Hospital General Universitario of Valencia, Spain
Birr, Katja	Department of Anesthesiology and Intensive Care, Pulmonary Engineering Group, University Hospital Carl Gustav Carus, Technische Universität Dresden, Germany
Bluth, Thomas	Department of Anesthesiology and Intensive Care, Pulmonary Engineering Group, University Hospital Carl Gustav Carus, Technische Universität Dresden, Germany
Bojaxhi, Elird	Department of Anesthesiology and Perioperative Medicine, Mayo Clinic Jacksonville, USA
Bonenti, Chiara	Università dell'Insubria – Dipartimento di Anestesia - Azienda Ospedaliera Asst Settlaghi Ospedale di Circolo e Fondazione Macchi – 21100 Varese - Italy
Bonney, Iwona	Department of Anesthesiology and Perioperative Medicine, Tufts Medical Center, Boston, MA, USA
Bos, Elke M.E.	Amsterdam UMC, University of Amsterdam, Department of Anesthesiology, Meibergdreef 9, Amsterdam, The Netherlands
Bowman, Sara	Homerton University Hospitals NHS Foundation Trust, UK
Braz, Leandro Gobbo	Department of Anesthesiology, Universidade Estadual Paulista, Botucatu/SP, Brazil
Brugnoni, Elisa	Università dell'Insubria – Dipartimento di Anestesia - Azienda Ospedaliera Asst Settlaghi Ospedale di Circolo e Fondazione Macchi – 21100 Varese - Italy
Brull, Sorin J	Department of Anesthesiology and Perioperative Medicine, College of Medicine, Mayo Clinic Jacksonville, FL, USA
Brunetti, Iole	Anestesia e Terapia Intensiva - IRCCS Policlinico San Martino – Genova – Italy
Bruni, Andrea	Anesthesia and Intensive Care, Department of Medical and Surgical Sciences, Magna Graeca University, Catanzaro, Italy
Buenvenida, Shonie L.	Department of Anesthesiology and Perioperative Medicine, Mayo Clinic, Rochester, Minnesota, USA
Busch, Cornelius Johannes*	Department of Anesthesiology, Heidelberg University Hospital, Im Neuenheimer Feld 110, 69120 Heidelberg, Germany
Camerini, Giovanni	Dipartimento di Scienze Chirurgiche e Diagnostiche Integrate – DISC – Università degli Studi di Genova – Genova – Italy and Clinica Chirurgica 1 - IRCCS Policlinico San Martino – Genova – Italy

Canet, Jaume*	University Hospital Germans Trias i Pujol, ICS, UAB, Spain
Capatti, Beatrice	Department Morphology, Surgery and Experimental Medicine. Anesthesia and intensive care university section. University of Ferrara. Italy
Carmona, Javiera	Department of Anesthesiology. School of Medicine. Pontificia Universidad Catolica de Chile
Carungcong, Jaime	Magill Department of Anaesthesia, Chelsea & Westminster NHS Foundation Trust, 369 Fulham Road, London, SW10 9NH, UK
Carvalho, Marta	Serviço de Anestesiologia, Centro Hospitalar do Porto, Porto, Portugal
Cattan, Anat	Division of Anesthesia, Pain and Critical Care, Tel-Aviv Medical Center affiliated with Sackler Medical School, Tel Aviv University, Tel Aviv, Israel
Cavaleiro, Carla	Serviço de Anestesiologia, Centro Hospitalar do Porto, Porto, Portugal and Centro de Investigação Clínica em Anestesiologia, Serviço de Anestesiologia, Centro Hospitalar do Porto, Porto, Portugal
Chiumello, Davide*	Dipartimento di Scienze della Salute, Università degli Studi di Milano, Milan, Italy and SC Anestesia e Rianimazione, ASST Santi Paolo e Carlo, Milan, Italy, Centro ricerca coordinata insufficienza respiratoria
Ciarlo, Stefano	Department Morphology, Surgery and Experimental Medicine. Anesthesia and intensive care university section. University of Ferrara. Italy
Cinella, Gilda	Department of Anesthesiology and Intensive Care Medicine, University of Foggia, Italy
Coburn, Mark	Department of Anesthesiology, Medical Faculty, RWTH Aachen University, Aachen, Germany
Colella, Umberto	Department of Women, child and General and Specialized Surgery. University of Campania "L. Vanvitelli", Italy.
Contreras, Victor	Department of Anesthesiology. School of Medicine. Pontificia Universidad Catolica de Chile
Corman Dincer, Pelin*	Marmara University Pendik Training and Research Hospital, Turkey
Cotter, Elizabeth	Section of Critical Care Medicine, Department of Anesthesia and Critical Care, University of Chicago, Chicago, IL, USA
Crovetto, Marcia	Department of Anesthesiology. School of Medicine. Pontificia Universidad Catolica de Chile
Darrah, William	Department of Anesthesiology, St. Michael's Hospital, University of Toronto, Toronto, ON, Canada
Davies, Simon*	York Teaching Hospitals NHS Foundation Trust, UK
de Baerdemaeker, Luc*	Ghent University Hospital, Ghent, Belgium. Corneel Heymanslaan 10, 9000 Gent, Belgium
De Hert, Stefan	Ghent University Hospital, Ghent, Belgium. Corneel Heymanslaan 10, 9000 Gent, Belgium
Del Cojo Peces, Enrique	Área de Salud Don Benito-Villanueva/Don Benito-Villanueva Health Area, Spain.
Delphin, Ellise	Montefiore Medical Center, Bronx, NY 10467, USA
Diaper, John	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland

do Nascimento Junior, Paulo*	Department of Anesthesiology, Universidade Estadual Paulista, Botucatu/SP, Brazil
Donatiello, Valerio	Department of Women, child and General and Specialized Surgery. University of Campania "L. Vanvitelli", Italy.
Dong, Jing	Department of Anesthesiology, Fudan University Shanghai Cancer Center, Shanghai, China. Department of Oncology, Shanghai Medical College, Fudan University, Shanghai, China
Dourado, Maria do Socorro	Associação Hospitalar Beneficente São Vicente de Paulo, Brazil
Dullenkopf, Alexander*	Frauenfeld Cantonal Hospital, Switzerland
Ebner, Felix	Klinik für Anästhesiologie, Intensivmedizin und Schmerztherapie, Universitätsklinikum Knappschafts Krankenhaus Bochum, Bochum, Germany
Elgendy, Hamed*	Departments of Anesthesiology, King Abdullah Medical City, Makkah, Saudi Arabia and Assiut University, Egypt and Hamad Medical Corporation, Qatar.
Ellenberger, Christoph	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Erdoğan Arı, Dilek*	Fatih Sultan Mehmet Educational and Research Hospital, İstanbul, Turkey
Ermert, Thomas	Department of Anesthesiology, Intensive Care and Pain Medicine, University Hospital Münster, Münster, Germany
Farah, Fadi	St. Elizabeth's Medical Center, MA, USA
Fernandez-Bustamante, Ana*	University of Colorado School of Medicine, Department of Anesthesiology, CO, USA
Ferreira, Cristina	Serviço de Anestesiologia, Centro Hospitalar do Porto, Porto, Portugal
Fiore, Marco	Department of Women, child and General and Specialized Surgery. University of Campania "L. Vanvitelli", Italy.
Fonte, Ana	Department of Anesthesiology from Centro Hospitalar de Entre o Douro e Vouga, EPE – São Sebastião Hospital, Santa Maria da Feira, Portugal
Fortià Palahí, Christina	Hospital Sagrat Cor- Barcelona, Spain
Galimberti, Andrea	Dipartimento di Scienze della Salute, Università degli Studi di Milano, Milan, Italy and SC Anestesia e Rianimazione, Asst Santi Paolo e Carlo, Milan, Italy, Centro ricerca coordinata insufficienza respiratoria
Gama de Abreu, Marcelo*	Department of Anesthesiology and Intensive Care, Pulmonary Engineering Group, University Hospital Carl Gustav Carus, Technische Universität Dresden, Germany
Garofano, Najia	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Giacconi, Luca Gregorio	Department of Women, child and General and Specialized Surgery. University of Campania "L. Vanvitelli", Italy.
Gilsanz, Fernando	Department of Anesthesiology and Surgical Critical Care, Hospital Universitario La Paz, Madrid, Spain
Girrbach, Felix	Department of Anaesthesiology and Intensive Care Medicine, University of Leipzig, Leipzig, Germany;
Gobbi, Luca	Department Morphology, Surgery and Experimental Medicine. Anesthesia and intensive care university section. University of Ferrara. Italy

Godfried, Marc Bernard*	Department of Anesthesiology, Onze Lieve Vrouwe Gasthuis, Amsterdam, The Netherlands
Goettel, Nicolai*	Department of Anesthesia, Surgical Intensive Care, Prehospital Emergency Medicine and Pain Therapy, University Hospital Basel, University of Basel, Basel, Switzerland
Goldstein, Peter A.*	Weill Cornell Medicine, New York, NY, USA
Goren, Or	Division of Anesthesia, Pain and Critical Care, Tel-Aviv Medical Center affiliated with Sackler Medical School, Tel Aviv University, Tel Aviv, Israel
Gorlin, Andrew	Mayo Clinic Arizona, 5777 East Mayo Blvd, Phoenix, AZ, 85054, USA
Granell Gil, Manuel*	Consorcio Hospital General Universitario of Valencia, Spain
Gratarola, Angelo	Anestesia e Rianimazione - IRCCS Policlinico San Martino – Genova – Italy
Graterol, Juan*	Royal Cornwall Hospital NHS Trust, UK
Guyon, Pierre	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Haire, Kevin	Magill Department of Anaesthesia, Chelsea & Westminster NHS Foundation Trust, 369 Fulham Road, London, SW10 9NH, UK
Harou, Philippe	Polyclinique Montier la Celle, France
Helf, Antonia	Department of Anaesthesiology, University Hospital Würzburg, Würzburg, Germany
Hemmes, Sabrina N. T.*	Amsterdam UMC, University of Amsterdam, Department of Anesthesiology, Meibergdreef 9, Amsterdam, The Netherlands
Hempel, Gunther	Department of Anaesthesiology and Intensive Care Medicine, University of Leipzig, Leipzig, Germany;
Hernández Cádiz, María José	Consorcio Hospital General Universitario of Valencia, Spain
Heyse, Björn	Ghent University Hospital, Ghent, Belgium. Corneel Heymanslaan 10, 9000 Gent, Belgium
Hollmann, Markus W.*	Amsterdam UMC, University of Amsterdam, Department of Anesthesiology, Meibergdreef 9, Amsterdam, The Netherlands
Huercio, Ivan	Department of Anesthesiology and Surgical Critical Care, Hospital Universitario La Paz, Madrid, Spain
Ilievska, Jasmina	University Clinic of Surgery “Ss. Naum Ohridski”, Faculty of Medicine, University “Ss. Cyril and Methodius”, Skopje, Macedonia
Jakus, Lien	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Jeganath, Vijay	Royal Stoke University Hospital, UHNM NHS Trust, Stoke-on-Trent, Staffordshire, ST4 6QG, UK
Jelting, Yvonne	Department of Anaesthesiology, University Hospital Würzburg, Würzburg, Germany
Jung, Minoa	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Kabon, Barbara*	Department of Anaesthesia, Critical Care and Pain Medicine, Medical University Vienna, Austria
Kacha, Aalok*	Section of Critical Care Medicine, Department of Anesthesia and Critical Care, University of Chicago, Chicago, IL, USA
Karaman Ilić, Maja	Clinical hospital Sveti Duh, JJ Strossmayer, Faculty of Medicine, Zagreb, Croatia

Karuppiah, Arunthevaraja Kavas, Ayse Duygu	St. Elizabeth's Medical Center, MA, USA Marmara University Pendik Training and Research Hospital, Turkey
Keli Barcelos, Gleicy	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Kellogg, Todd A.	Department of Surgery, Mayo Clinic, Rochester, Minnesota, USA
Kemper, Johann	Department of Anesthesiology, University Hospital Duesseldorf, Heinrich-Heine University Duesseldorf, Duesseldorf, Germany
Kerbrat, Romain	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Khodr, Suraya Kienbaum, Peter	Cleveland Clinic Abu Dhabi, Al Maryah Island, Abu Dhabi, UAE Department of Anesthesiology, University Hospital Duesseldorf, Heinrich-Heine University Duesseldorf, Duesseldorf, Germany
Kir, Bunyamin	Marmara University Pendik Training and Research Hospital, Turkey
Kiss, Thomas	Department of Anesthesiology and Intensive Care, Pulmonary Engineering Group, University Hospital Carl Gustav Carus, Technische Universität Dresden, Germany
Kivrak, Selin	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Klarić, Vlasta*	Clinic of anesthesiology, reanimatology and intensive care medicine, University Hospital Dubrava, Zagreb, Croatia
Koch, Thea	Department of Anesthesiology and Intensive Care, Pulmonary Engineering Group, University Hospital Carl Gustav Carus, Technische Universität Dresden, Germany
Köksal, Ceren	Fatih Sultan Mehmet Educational and Research Hospital, İstanbul, Turkey
Kowark, Ana	Department of Anesthesiology, Medical Faculty, RWTH Aachen University, Aachen, Germany
Kranke, Peter*	Department of Anaesthesiology, University Hospital Würzburg, Würzburg, Germany
Kuvaki, Bahar*	Dokuz Eylül University Faculty of Medicine, Department of Anaesthesiology and Intensive Care, Turkey
Kuzmanovska, Biljana	University clinic for Traumatology, Orthopedics, Anesthesia, Reanimation, Intensive care and Emergency center - Skopje, Department of Anesthesiology, Reanimation and Intensive care Medicine - Skopje, Macedonia
Laffey, John*	Department of Anesthesiology, St. Michael's Hospital, University of Toronto, Toronto, ON, Canada
Lange, Mirko	Department of Anaesthesiology and Intensive Care Medicine, University of Leipzig, Leipzig, Germany;
Lemos, Marília Freitas de	Department of Anesthesiology, Hospital do Câncer II, National Cancer Institute of Brazil (INCA), Rio de Janeiro, Brazil
Licker, Marc-Joseph*	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie; Faculty of Medicine, University of Geneva, Switzerland
López-Baamonde, Manuel	Department of Anaesthesiology, Hospital Clínic, Barcelona, Spain.

López-Hernández, Antonio	Department of Anaesthesiology, Hospital Clínic, Barcelona, Spain.
Lopez-Martinez, Mercedes	Department of Anesthesiology and Surgical Critical Care, Hospital Universitario La Paz, Madrid, Spain
Luise, Stéphane	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
MacGregor, Mark*	Ashford and St Peters NHS Foundation Trust, UK
Machado, Humberto S.*	Serviço de Anestesiologia, Centro Hospitalar do Porto, Porto, Portugal and Instituto Ciências Biomédicas Abel Salazar, Universidade do Porto, Porto, Portugal and Centro de Investigação Clínica em Anestesiologia, Serviço de Anestesiologia, Centro Hospitalar do Porto, Porto, Portugal
Magalhães, Danielle*	Associação Hospitalar Beneficente São Vicente de Paulo, Brazil
Maillard, Julien	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Malerbi, Patrizia	Anestesia e Rianimazione - IRCCS Policlinico San Martino – Genova – Italy
Manimekalai, Natesan	University of Mississippi Medical Center, MS, USA
Margarson, Michael*	St Richard's Hospital, Chichester, UK
Markstaller, Klaus*	Department of Anaesthesia, Critical Care and Pain Medicine, Medical University Vienna, Austria
Martin, Archer K.	Department of Anesthesiology and Perioperative Medicine, College of Medicine, Mayo Clinic Jacksonville, FL, USA
Martin, David P.	Department of Anesthesiology and Perioperative Medicine, Mayo Clinic, Rochester, Minnesota, USA
Martin, Yvette N.	Department of Anesthesiology and Perioperative Medicine, Mayo Clinic, Rochester, Minnesota, USA
Martínez-Ocon, Julia	Department of Anaesthesiology, Hospital Clínic, Barcelona, Spain.
Martin-Loeches, Ignacio*	Department of Intensive Care Medicine, Multidisciplinary Intensive Care Research Organization (MICRO), St. James's Hospital, Dublin, Ireland
Maseda, Emilio	Department of Anesthesiology and Surgical Critical Care, Hospital Universitario La Paz, Madrid, Spain
Matot, Idit*	Division of Anesthesia, Pain and Critical Care, Tel-Aviv Medical Center affiliated with Sackler Medical School, Tel Aviv University, Tel Aviv, Israel
McAuliffe, Niamh	Department of Anesthesiology, St. Michael's Hospital, University of Toronto, Toronto, ON, Canada
McKenzie, Travis J.	Department of Surgery, Mayo Clinic, Rochester, Minnesota, USA
Medina, Paulina	Department of Anesthesiology. School of Medicine. Pontificia Universidad Catolica de Chile
Meersch, Melanie	Department of Anesthesiology, Intensive Care and Pain Medicine, University Hospital Münster, Münster, Germany
Menzen, Angelika*	St. Marienhospital gGmbH Friesoythe, Friesoythe, Germany
Mertens, Els*	University Hospital Antwerp, Belgium
Meurer, Bernd	Marienhospital Wesel, Wesel, Germany
Meyer-Treschan, Tanja*	Department of Anesthesiology, University Hospital Duesseldorf, Heinrich-Heine University Duesseldorf, Duesseldorf, Germany

Miao, Changhong*	Department of Anesthesiology, Fudan University Shanghai Cancer Center, Shanghai, China. Department of Oncology, Shanghai Medical College, Fudan University, Shanghai, China
Micalizzi, Camilla	Dipartimento di Scienze Chirurgiche e Diagnostiche Integrate – DISC – Università degli Studi di Genova – Genova – Italy and Anestesia e Terapia Intensiva - IRCCS Policlinico San Martino – Genova – Italy
Milić, Morena	Clinic of anesthesiology, reanimatology and intensive care medicine, University Hospital Dubrava, Zagreb, Croatia
Módoło, Norma Sueli Pinheiro	Department of Anesthesiology, Universidade Estadual Paulista, Botucatu/SP, Brazil
Moine, Pierre	University of Colorado School of Medicine, Department of Anesthesiology, CO, USA
Mölders, Patrick	Klinik für Anästhesiologie, Intensivmedizin und Schmerztherapie, Universitätsklinikum Knappschafts Krankenhaus Bochum, Bochum, Germany
Montero-Feijoo, Ana	Department of Anesthesiology and Surgical Critical Care, Hospital Universitario La Paz, Madrid, Spain
Moret, Enrique	University Hospital Germans Trias i Pujol, ICS, UAB, Spain
Muller, Markus K.	Frauenfeld Cantonal Hospital, Switzerland
Murphy, Zoe	York Teaching Hospitals NHS Foundation Trust, UK
Nalwaya, Pramod*	Royal Stoke University Hospital, UHNH NHS Trust, Stoke-on-Trent, Staffordshire, ST4 6QG, UK
Naumovski, Filip*	University clinic for Traumatology, Orthopedics, Anesthesia, Reanimation, Intensive care and Emergency center - Skopje, Department of Anesthesiology, Reanimation and Intensive care Medicine - Skopje, Macedonia
Navalesi, Paolo*	Anesthesia and Intensive Care, Department of Medical and Surgical Sciences, Magna Graeca University, Catanzaro, Italy
Navarro e Lima, Lais Helena	Department of Anesthesiology, Universidade Estadual Paulista, Botucatu/SP, Brazil
Nesek Adam, Višnja*	Clinical hospital Sveti Duh, JJ Strossmayer, Faculty of Medicine, Zagreb, Croatia
Neumann, Claudia	Department of Anaesthesiology and Intensive Care Medicine, University Hospital of Bonn, Bonn, Germany
Newell, Christopher*	Southmead Hospital, North Bristol NHS Trust, UK
Nisnevitch, Zoulfira*	Montefiore Medical Center, Bronx, NY 10467, USA
Nizamuddin, Junaid	Section of Critical Care Medicine, Department of Anesthesia and Critical Care, University of Chicago, Chicago, IL, USA
Novazzi, Cecilia	Università dell'Insubria – Dipartimento di Anestesia - Azienda Ospedaliera Asst Settlaghi Ospedale di Circolo e Fondazione Macchi – 21100 Varese - Italy
Nunes, Catarina S.	Universidade Aberta, Departamento de Ciências e Tecnologia, Porto, Portugal and Centro de Investigação Clínica em Anestesiologia, Serviço de Anestesiologia, Centro Hospitalar do Porto, Porto, Portugal
O'Connor, Michael	Section of Critical Care Medicine, Department of Anesthesia and Critical Care, University of Chicago, Chicago, IL, USA
Oprea, Günther	Klinik für Anästhesiologie, Intensivmedizin und Schmerztherapie, Universitätsklinikum Knappschafts Krankenhaus Bochum, Bochum, Germany

Orhan Sungur, Mukadder Özbilgin, Şule	Istanbul University, Istanbul Faculty of Medicine, Turkey Dokuz Eylül University Faculty of Medicine, Department of Anaesthesiology and Intensive Care, Turkey
Pace, Maria Caterina	Department of Women, child and General and Specialized Surgery. University of Campania "L. Vanvitelli", Italy.
Pacheco, Marcos	Department of Anesthesiology from Centro Hospitalar de Entre o Douro e Vouga, EPE – São Sebastião Hospital, Santa Maria da Feira, Portugal
Packianathaswamy, Balaji*	Hull and East Yorkshire Hospitals NHS Trust, UK
Palma Gonzalez, Estefania	Área de Salud Don Benito-Villanueva/Don Benito-Villanueva Health Area, Spain.
Papaspoulos, Fotios*	POLYCLINIQUE MONTIER LA CELLE, France
Paredes, Sebastián	Department of Anesthesiology. School of Medicine. Pontificia Universidad Catolica de Chile
Passavanti, Maria Beatrice	Department of Women, child and General and Specialized Surgery. University of Campania "L. Vanvitelli", Italy.
Pedemonte, Juan Cristobal	Department of Anesthesiology. School of Medicine. Pontificia Universidad Catolica de Chile
Pelosi, Paolo*	Dipartimento di Scienze Chirurgiche e Diagnostiche Integrate – DISC – Università degli Studi di Genova – Genova – Italy and Anestesia e Terapia Intensiva - IRCCS Policlinico San Martino – Genova – Italy
Peremin, Sanja	Clinic of anesthesiology, reanimatology and intensive care medicine, University Hospital Dubrava, Zagreb, Croatia
Philipsenburger, Christoph	Department of Anesthesiology, Heidelberg University Hospital, Im Neuenheimer Feld 110, 69120 Heidelberg, Germany
Pinho, Daniela	Serviço de Anestesiologia, Centro Hospitalar do Porto, Porto, Portugal
Pinho, Silvia	Serviço de Anestesiologia, Centro Hospitalar do Porto, Porto, Portugal
Posthuma, Linda M.	Amsterdam UMC, University of Amsterdam, Department of Anesthesiology, Meibergdreef 9, Amsterdam, The Netherlands
Pota, Vincenzo	Department of Women, child and General and Specialized Surgery. University of Campania "L. Vanvitelli", Italy.
Preckel, Benedikt	Amsterdam UMC, University of Amsterdam, Department of Anesthesiology, Meibergdreef 9, Amsterdam, The Netherlands
Priani, Paolo	Department Morphology, Surgery and Experimental Medicine. Anesthesia and intensive care university section. University of Ferrara. Italy
Putensen, Christian*	Department of Anaesthesiology and Intensive Care Medicine, University Hospital of Bonn, Bonn, Germany
Rached, Mohamed Aymen	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Radoeshki, Aleksandar	University Clinic of Surgery "Ss. Naum Ohridski", Faculty of Medicine, University "Ss. Cyril and Methodius", Skopje, Macedonia
Ragazzi, Riccardo	Department Morphology, Surgery and Experimental Medicine. Anesthesia and intensive care university section. University of Ferrara. Italy

Rajamanickam, Tamilselvan Rajamohan, Arthi	Royal Stoke University Hospital, UHNM NHS Trust, Stoke-on-Trent, Staffordshire, ST4 6QG, UK Department of Anesthesiology, St. Michael's Hospital, University of Toronto, Toronto, ON, Canada
Ramakrishna, Harish*	Mayo Clinic Arizona, 5777 East Mayo Blvd, Phoenix, AZ, 85054, USA
Rangarajan, Desikan* Reiterer, Christian	Homerton University Hospitals NHS Foundation Trust, UK Department of Anaesthesia, Critical Care and Pain Medicine, Medical University Vienna, Austria
Renew, J. Ross	Department of Anesthesiology and Perioperative Medicine, Mayo Clinic Jacksonville, USA
Reynaud, Thomas	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Rhys, Rhidian Rivas, Eva*	Southmead Hospital, North Bristol NHS Trust, UK Department of Anaesthesiology, Hospital Clínic, IDIBAPS, Universitat de Barcelona, Spain
Robitzky, Luisa	Klinik für Anästhesiologie, Intensivmedizin und Schmerztherapie, Universitätsklinikum Knappschafts Krankenhaus Bochum, Bochum, Germany
Rossaint, Rolf*	Department of Anesthesiology, Medical Faculty, RWTH Aachen University, Aachen, Germany
Rubulotta, Francesca*	Imperial College Healthcare NHS Trust, UK
Sabbatini, Giovanni	Dipartimento di Scienze della Salute, Università degli Studi di Milano, Milan, Italy and SC Anestesia e Rianimazione, ASST Santi Paolo e Carlo, Milan, Italy, Centro ricerca coordinata insufficienza respiratoria
Samuels, Jon D.	Weill Cornell Medicine, Department of Anesthesiology New York-Presbyterian Hospital, New York, NY, USA
Sanahuja, Josep Martí	Department of Anaesthesiology, Hospital Clínic, Barcelona, Spain.
Sansone, Pasquale	Department of Women, child and General and Specialized Surgery. University of Campania "L. Vanvitelli", Italy.
Santos, Alice	Department of Anaesthesiology, Centro Hospitalar São João, Porto, Portugal
Sayedahl, Mohamed	Departments of Anesthesiology, King Abdullah Medical City, Makkah, Saudi Arabia and Mansoura University, Egypt
Schaefer, Maximilian S.	Department of Anesthesiology, University Hospital Duesseldorf, Heinrich-Heine University Duesseldorf, Duesseldorf, Germany
Scharffenberg, Martin	Department of Anesthesiology and Intensive Care, Pulmonary Engineering Group, University Hospital Carl Gustav Carus, Technische Universität Dresden, Germany
Schiffer, Eduardo	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie; Faculty of Medicine, University of Geneva, Switzerland
Schliewe, Nadja	Department of Anaesthesiology and Intensive Care Medicine, University of Leipzig, Leipzig, Germany;
Schorer, Raoul	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Schultz, Marcus J.	Amsterdam UMC, University of Amsterdam, Department of Anesthesiology, Meibergdreef 9, Amsterdam, The Netherlands

Schumann, Roman	Department of Anesthesiology and Perioperative Medicine, Tufts Medical Center, Boston, MA, USA
Selmo, Gabriele	Universita' dell'Insubria – Dipartimento di Anestesia - Azienda Ospedaliera Asst Settlaghi Ospedale di Circolo e Fondazione Macchi – 21100 Varese - Italy
Sendra, Mar	University Hospital Germans Trias i Pujol, ICS, UAB, Spain
Senturk, Mert*	Istanbul University, Istanbul Faculty of Medicine, Turkey
Severgnini, Paolo*	Universita' dell'Insubria – Dipartimento di Anestesia - Azienda Ospedaliera ASST Sette Laghi Ospedale di Circolo e Fondazione Macchi Varese, Italy
Shaw, Kate	York Teaching Hospitals NHS Foundation Trust, UK
Shosholcheva, Mirjana*	University Clinic of Surgery “Ss. Naum Ohridski”, Faculty of Medicine, University “Ss. Cyril and Methodius”, Skopje, Macedonia
Sibai, Abdulrazak*	MINISTRY OF NATIONAL GUARD HEALTH AFFAIRS, KING ABDULAZIZ MEDICAL CITY – RIYADH, Anesthesia Department, Saudi Arabia
Simon, Philipp	Department of Anaesthesiology and Intensive Care Medicine, University of Leipzig, Leipzig, Germany;
Simonassi, Francesca	Dipartimento di Scienze Chirurgiche e Diagnostiche Integrate – DISC – Università degli Studi di Genova – Genova – Italy and Anestesia e Terapia Intensiva - IRCCS Policlinico San Martino – Genova – Italy
Sinno, Claudia	Universita' dell'Insubria – Dipartimento di Anestesia - Azienda Ospedaliera Asst Settlaghi Ospedale di Circolo e Fondazione Macchi – 21100 Varese - Italy
Sivriköz, Nukhet	Istanbul University, Istanbul Faculty of Medicine, Turkey
Skandalou, Vasiliki*	Alexandra General Hospital, Athens, Greece
Smith, Neil	Hull and East Yorkshire Hospitals NHS Trust, UK
Soares, Maria	Serviço de Anestesiologia, Centro Hospitalar do Porto, Porto, Portugal
Socorro Artilles, Tania	University Clinic Hospital Valencia, Spain
Sousa Castro, Diogo*	Department of Anesthesiology from Centro Hospitalar de Entre o Douro e Vouga, EPE – São Sebastião Hospital, Santa Maria da Feira, Portugal
Sousa, Miguel	Department of Anesthesiology from Centro Hospitalar de Entre o Douro e Vouga, EPE – São Sebastião Hospital, Santa Maria da Feira, Portugal
Spadaro, Savino*	Department Morphology, Surgery and Experimental Medicine. Anesthesia and intensive care university section. University of Ferrara. Italy
Sprung, Juraj*	Department of Anesthesiology and Perioperative Medicine, Mayo Clinic, Rochester, Minnesota, USA
Stamatakis, Emmanouil	Alexandra General Hospital, Athens, Greece
Steiner, Luzius A.	Department of Anesthesia, Surgical Intensive Care, Prehospital Emergency Medicine and Pain Therapy, University Hospital Basel, University of Basel, Basel, Switzerland
Stevenazzi, Andrea	University Hospital Germans Trias i Pujol, ICS, UAB, Spain
Suarez-de-la-Rica, Alejandro*	Department of Anesthesiology and Surgical Critical Care, Hospital Universitario La Paz, Madrid, Spain

Suppan, Mélanie	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Teichmann, Robert	Department of Anesthesiology and Intensive Care, Pulmonary Engineering Group, University Hospital Carl Gustav Carus, Technische Universität Dresden, Germany
Tena Guerrero, José Maria*	Área de Salud Don Benito-Villanueva/Don Benito-Villanueva Health Area, Spain.
Thiel, Bram	Department of Anesthesiology, Onze Lieve Vrouwe Gasthuis, Amsterdam, The Netherlands
Tolós, Raquel	University Hospital Germans Trias i Pujol, ICS, UAB, Spain
Tore Altun, Gulbin	Marmara University Pendik Training and Research Hospital, Turkey
Tucci, Michelle	University of Mississippi Medical Center, MS, USA
Turnbull, Zachary A.	Weill Cornell Medicine, New York, NY, USA
Turudić, Žana	Clinic of anesthesiology, reanimatology and intensive care medicine, University Hospital Dubrava, Zagreb, Croatia
Unterberg, Matthias	Klinik für Anästhesiologie, Intensivmedizin und Schmerztherapie, Universitätsklinikum Knappschaftskrankenhaus Bochum, Bochum, Germany
Van Limmen, Jurgen	Ghent University Hospital, Ghent, Belgium. Corneel Heymanslaan 10, 9000 Gent, Belgium
Van Nieuwenhove, Yves	Ghent University Hospital, Ghent, Belgium. Corneel Heymanslaan 10, 9000 Gent, Belgium
Van Waesberghe, Julia	Department of Anesthesiology, Medical Faculty, RWTH Aachen University, Aachen, Germany
Vidal Melo, Marcos Francisco*	Department of Anesthesia, Critical Care and Pain Medicine, Massachusetts General Hospital, Boston, Massachusetts, USA
Vitković, Bibiana	Clinic of anesthesiology, reanimatology and intensive care medicine, University Hospital Dubrava, Zagreb, Croatia
Vivona, Luigi	Department of Anesthesiology and Intensive Care, Pulmonary Engineering Group, University Hospital Carl Gustav Carus, Technische Universität Dresden, Germany
Vizcaychipi, Marcela*	Magill Department of Anaesthesia, Chelsea & Westminster NHS Foundation Trust, 369 Fulham Road, London, SW10 9NH, UK
Volta, Carlo Alberto	Department Morphology, Surgery and Experimental Medicine. Anesthesia and intensive care university section. University of Ferrara. Italy
Weber, Anne	Hôpitaux Universitaires de Genève, Département APSI, service d'anesthésiologie, Switzerland
Weingarten, Toby N.	Department of Anesthesiology and Perioperative Medicine, Mayo Clinic, Rochester, Minnesota, USA
Wittenstein, Jakob	Department of Anesthesiology and Intensive Care, Pulmonary Engineering Group, University Hospital Carl Gustav Carus, Technische Universität Dresden, Germany
Wrigge, Hermann*	Department of Anaesthesiology and Intensive Care Medicine, University of Leipzig, Leipzig, Germany and Department of Anaesthesiology, Intensive Care and Emergency Medicine, Pain Therapy; Bergmannstrost Hospital Halle, Halle, Germany
Wyffels, Piet	Ghent University Hospital, Ghent, Belgium. Corneel Heymanslaan 10, 9000 Gent, Belgium

Yagüe, Julio	Department of Anesthesiology and Surgical Critical Care, Hospital Universitario La Paz, Madrid, Spain
Yates, David	York Teaching Hospitals NHS Foundation Trust, UK
Yavru, Ayşen	Istanbul University, Istanbul Faculty of Medicine, Turkey
Zac, Lilach	Division of Anesthesia, Pain and Critical Care, Tel-Aviv Medical Center affiliated with Sackler Medical School, Tel Aviv University, Tel Aviv, Israel
Zhong, Jing	Department of Anesthesiology, Fudan University Shanghai Cancer Center, Shanghai, China. Department of Oncology, Shanghai Medical College, Fudan University, Shanghai, China

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eMethods

2.1 Inclusion and exclusion criteria

Patients were included if they had a body mass index (BMI) ≥ 35 kg/m², were scheduled for laparoscopic or non-laparoscopic surgery expected to exceed 2h (from incision to closure) under general anesthesia, and had an intermediate to high risk of developing postoperative pulmonary complications according to the Assess Respiratory Risk in Surgical Patients in Catalonia (ARISCAT) score ¹ (**Table S1**). This score predicts individual preoperative risk for postoperative pulmonary complications using 7 independent predictors, of which 4 are patient-related and 3 surgery-related predictors. An ARISCAT risk score ≥ 26 is associated with an intermediate-to-high risk for postoperative pulmonary complications.

Patients were excluded from participation if they: were aged <18 years; underwent any kind of previous lung surgery; were invasively mechanically ventilated for longer than 30 minutes within the last 30 days before surgery; or they received recent immunosuppressive medication (chemotherapy or radiation therapy up to two months prior to surgery). Further exclusion criteria comprises neurosurgical procedures and cardiac surgery, need for one-lung ventilation or planned re-intubation following surgery, need for intraoperative prone or lateral decubitus position and enrollment in another interventional study or refusal to give written informed consent. Additionally, patients showing at least one the following medical conditions are excluded: pregnancy (excluded by anamnesis and /or laboratory analysis), persistent hemodynamic instability or intractable shock (considered hemodynamically unsuitable for the study by the patient's managing physician), history of severe chronic obstructive pulmonary disease (COPD, defined as non-invasive ventilation and/or oxygen therapy at home or repeated systemic corticosteroid therapy for acute exacerbations of COPD), severe cardiac disease (defined as New York Heart Association class III or IV, acute coronary syndrome or persistent ventricular tachyarrhythmia), concurrent acute respiratory distress syndrome (according to the Berlin definition ²) expected to require prolonged postoperative mechanical ventilation, severe pulmonary arterial hypertension (defined as systolic pulmonary artery pressure > 40 mmHg), intracranial injury or tumor or neuromuscular disease.

2.2 Description of the trial interventions

Patients underwent intraoperative lung protective mechanical ventilation with protective low tidal volume of 7 mL/kg predicted body weight (PBW), and were randomly assigned to a PEEP level of 12 cmH₂O with planned lung recruitment maneuvers performed after intubation, hourly thereafter and preceding extubation (high PEEP group), or a level of PEEP of 4 cmH₂O without planned recruitment maneuvers (low PEEP group). PEEP levels were maintained throughout the whole period of intraoperative mechanical ventilation.

2.3 Standard procedures and recommendations to anesthesiologists

To avoid interference with the trial intervention, routine elements of perioperative anesthesia care (including general anesthesia, postoperative pain management, physiotherapeutic procedures and fluid management) were performed according to each site's specific expertise and clinical routine. The following approaches were recommended (not mandatory) for anesthetic management: 1) to use inhalational isoflurane, desflurane or sevoflurane as the volatile agent, intravenous propofol, remifentanyl or sufentanyl, and cis-atracurium, atracurium, vecuronium, or rocuronium as required; 2) to use balanced solution of prostigmine, or

neostigmine and atropine or glycopyrrolate for reversal of muscle relaxation, guided by neuromuscular function monitoring, 3) to perform postoperative pain management in order to achieve a visual analogue scale (VAS) pain score below 3 and regional or neuraxial analgesia should be used whenever indicated, 4) to use physiotherapy by early mobilization, deep breathing exercises with and without incentive spirometry and stimulation of cough in the postoperative period, 5) to avoid hypo- and hypervolemia, 6) to use invasive measurement of arterial blood pressure whenever indicated, 7) to use appropriate prophylactic antibiotic drugs whenever indicated.

In addition, the study protocol stressed that routine intraoperative monitoring should include measurements of noninvasive blood pressure, pulse oximetry, end-tidal carbon dioxide fraction and electrocardiography. Every patient should receive at least one peripheral venous line to allow adequate fluid resuscitation during the study period. Nasogastric tubes, urinary bladder catheters and/or other intravenous catheters, as well as other, more invasive monitoring may be used according to local practice and/or guidelines. Other procedures should follow the Safe Surgery Checklist of the World Health Organization as published at www.who.int/patientsafety/safesurgery/en/index.html.

2.4 Mechanical ventilation

Mechanical ventilation was performed with anesthesia ventilators in use in each individual site participating in the study. Patients were volume-controlled mechanically ventilated with the lowest possible FiO₂, but at least 0.4, to maintain SpO₂ of > 92%, an inspiratory to expiratory ratio (I:E) of 1:2 and a respiratory rate adjusted to normocapnia (end-tidal carbon dioxide partial pressure between 35 and 45 mmHg). It was left to the discretion of the attending anesthesiologist to use a higher FiO₂.

Tidal volume was set to 7 mL/kg PBW. The PBW was calculated according to a predefined formula: 50 + 0.91 x (centimeters of height – 152.4) for males, and 45.5 + 0.91 x (centimeters of height – 152.4) for females. Tidal volume throughout this protocol referred to the actual inspired tidal volume in the ventilator circuit.

PEEP was set according to the randomized intervention to 4 vs. 12 cmH₂O and only modified as part of the rescue strategy (in case of desaturation, see below), or at discretion of the treating physician.

2.5 Planned and unplanned recruitment maneuvers

The recruitment maneuver, as part of the high PEEP group strategy, was performed directly after induction of anesthesia, after any disconnection from the mechanical ventilator, every one hour during surgery and before extubation, in a hemodynamically stable situation as judged by the anesthesiologist. Recruitment maneuvers should also be performed as part of a rescue strategy in the low PEEP group. To obtain standardization among centers, recruitment maneuvers were performed in volume-controlled ventilation, according to the following steps:

- I. Peak inspiratory pressure limit set at 55 cmH₂O
- II. Tidal volume set at 7 mL/kg PBW and respiratory rate at 6 or higher breaths/min, while PEEP was 12 cmH₂O (or higher if during rescue, see below)
- III. I:E at 1:1
- IV. Tidal volume increased in steps of 4 mL/kg PBW until plateau pressure reached 40 – 50 cmH₂O

V. If the maximum tidal volume allowed by the anesthesia ventilator was achieved and the plateau pressure was lower than 40 cmH₂O, PEEP was increased as needed, but the maximum was 20 cmH₂O

VI. Three breaths were allowed while maintaining plateau pressure of 40 – 50 cmH₂O

VII. Respiratory rate, I:E, inspiratory pause and tidal volume were set back to pre-recruitment values, while maintaining PEEP at 12 cmH₂O (or higher if during rescue)

2.6 Rescue strategies for intraoperative hypoxemia

If SpO₂ ≤ 92% developed, increase in airway resistance, presence of intrinsic PEEP, hemodynamic impairment and ventilator malfunction must have been excluded before group-specific stepwise rescue strategies could be applied (**Table S2**).

In patients receiving lower PEEP levels, rescue consisted primarily of an increase in FiO₂, whereas elevation of PEEP levels was restricted to more severe cases of hypoxemia.

In the higher PEEP group, the rescue strategy consisted of primarily increase of PEEP before FiO₂ was increased. At any rescue step, the treating physician could consider reducing PEEP if SpO₂ deteriorated further in an otherwise hemodynamic stable patient.

2.7 Protocol deviation

Anesthesiologists could deviate from the ventilation protocol at any time if concerns about patient's safety arose, or upon the surgeon's request.

PEEP could be modified according to the anesthesiologist's judgment in presence of any of the following clinical situation: 1) decrease in systolic arterial pressure below 90 mmHg unresponsive to fluids and/or vasoactive drugs, 2) need for a dosage of vasoactive drugs at the tolerance limit, 3) new arrhythmias unresponsive to the treatment suggested by the Advanced Cardiac Life Support Guidelines ³, 4) blood loss requiring massive transfusion occurred (defined as replacement of >100% blood volume in 24 hours or >50% of blood volume in 4 hours to maintain hematocrit > 21% (hemoglobin > 7 mg/dL)) or 5) any life-threatening surgical complication occurred that might benefit from changes in PEEP.

2.8 Definitions

The components of the primary endpoint, namely a composite outcome of postoperative pulmonary complications were defined as follows: 1) mild respiratory failure (PaO₂ < 60 mmHg or SpO₂ < 90% breathing at least 10 min of room air but responding to supplemental oxygen = 2 L/min, excluding hypoventilation), 2) moderate respiratory failure (PaO₂ < 60 mmHg or SpO₂ < 90% breathing at least 10 min of room air but responding only to supplemental oxygen > 2 L/min, excluding hypoventilation), 3) severe respiratory failure (need for non-invasive or invasive mechanical ventilation, excluding hypoventilation due to sedative agents), 4) ARDS, 5) bronchospasm (newly detected expiratory wheezing treated with bronchodilators), 6) new pulmonary infiltrates (chest X-ray demonstrating new mono-lateral or bilateral infiltrate without other clinical signs), 7) pulmonary infection (new or progressive radiographic infiltrate plus at least two of the following: antibiotic treatment, tympanic temperature > 38°C, leukocytosis or leucopenia (white blood cell count < 4000 cells/mm³ or > 12000 cells/mm³) and/or purulent secretions), 8) aspiration pneumonitis (respiratory failure after the inhalation of regurgitated gastric contents), 9) pleural effusion (chest X-ray demonstrating blunting of the costophrenic angle, loss of the sharp silhouette of the ipsilateral hemidiaphragm in upright

position, evidence of displacement of adjacent anatomical structures, or (in supine position) a hazy opacity in one hemithorax with preserved vascular shadows), 10) atelectasis (lung opacification with shift of the mediastinum, hilum, or hemidiaphragm towards the affected area, and compensatory overinflation in the adjacent non-atelectatic lung), 11) cardiopulmonary edema (clinical signs of congestion, including dyspnea, edema, rales and jugular venous distention, with the chest X-ray demonstrating increase in vascular markings and diffuse alveolar interstitial infiltrates) and 12) pneumothorax (air in the pleural space with no vascular bed surrounding the visceral pleura).

The secondary outcomes were: 1) composite of severe postoperative pulmonary complications, defined as any of the above mentioned adverse pulmonary events, except to mild respiratory failure; 2) intraoperative adverse events, e.g. hypoxemia (defined as $\text{SpO}_2 \leq 92\%$), hypotension (defined as systolic blood pressure $< 90\text{mmHg}$) and bradycardia (defined as heart rate < 50 bpm), 3) unexpected need for intensive care unit (ICU) admission or ICU readmission, 4) hospital-free days at follow-up day 90, 5) impaired post-operative wound healing (defined as an interruption in the timely and predictable recovery of mechanical integrity in the injured tissue) and 6) 5-day and in-hospital mortality, as well as 7) postoperative extra-pulmonary complications (PEPCs). PEPCs included systemic inflammatory response syndrome (SIRS), sepsis, severe sepsis and septic shock (all according to consensus criteria ⁴), extra-pulmonary infection (wound infection or any other infection), coma (Glasgow Coma Score < 8 in the absence of therapeutic coma or sedation), acute myocardial infarction (according to universal definition of myocardial infarction ⁵), acute renal failure (according to the risk, injury, failure, loss, end-stage kidney disease [RIFLE] classification system ⁶), disseminated intravascular coagulation (DIC) (according to the International Society of Thrombosis and Hemostasis diagnostic scoring system for DIC ⁷), gastro-intestinal failure (GIF) (defined according to the GIF-score ⁸) and hepatic failure (defined as ratio of total bilirubin on postoperative day 5 to postoperative day 1 > 1.7 and ratio of international normalized ratio (INR) on postoperative day 5 to postoperative day 1 > 1.0 , or new presence of hepatic encephalopathy and coagulopathy (INR > 1.5) within 8 weeks after initial signs of liver injury (e.g. jaundice) without evidence for chronic liver disease)(adapted from Du et al. ⁹ and Wlodzimirow et al. ¹⁰).

eTable 1. The Assess Respiratory Risk in Surgical Patients in Catalonia (ARISCAT) score

	Multivariate Analysis, OR (95% CI), n = 1624	β Coefficient	Risk Score[§]
Age, years			
≤ 50	1		
51 – 80	1.4 (0.6 - 3.3)	0.331	3
> 80	5.1 (1.9 - 13.3)	1.619	16
Preoperative SpO₂, %			
≥ 96	1		
91 – 95	2.2 (1.2 - 4.2)	0.802	8
≤ 90	10.7 (4.1 - 28.1)	2.375	24
Respiratory infection in the last month	5.5 (2.6 - 11.5)	1.698	17
Preoperative anemia (≤ 10 g/dl)	3.0 (1.4 - 6.5)	1.105	11
Surgical incision			
Peripheral	1		
Upper abdominal	4.4 (2.3 - 8.5)	1.480	15
Intrathoracic	11.4 (4.9 - 26.0)	2.431	24
Duration of surgery, hours			
≤ 2	1		
> 2 to 3	4.9 (2.4 - 10.1)	1.593	16
> 3	9.7 (4.7 - 19.9)	2.268	23
Emergency procedure	2.2 (1.04 - 4.5)	0.768	8

Legend: A risk score ≥ 26 predicts an intermediate-to-high risk for postoperative pulmonary complications). CI, confidence interval; OR, odds ratio; SpO₂, peripheral oxyhemoglobin saturation by pulse oximetry breathing air in supine position; g/dL, gram per deciliter; § The simplified risk score was the sum of each logistic regression coefficient multiplied by 10, after rounding off its value

eTable 2. Rescue strategies for intraoperative hypoxemia

Step	Lower PEEP			Higher PEEP		
	FiO ₂	PEEP [cmH ₂ O]		FiO ₂	PEEP [cmH ₂ O]	
1	0.5	4		0.4	14	(+RM)
2	0.6	4		0.4	16	(+RM)
3	0.7	4		0.4	18	(+RM)
4	0.8	4		0.5	18	
5	0.9	4		0.6	18	
6	1.0	4		0.7	18	
7	1.0	5		0.8	18	
8	1.0	6		0.9	18	
9	1.0	7	(+RM)	1.0	18	
10				1.0	20	(+RM)

Legend: If intraoperative hypoxemia, defined as oxygen saturation $\leq 92\%$, develops, sequences of interventions will be used according to group assignment. FiO₂, inspiratory fraction of oxygen; PEEP, positive end-expiratory airway pressure; RM, lung recruitment maneuver

eTable 3. Types of intraoperative IV fluids

	High PEEP n=989	Low PEEP n=987	P value
Total fluids – mL, mean (SD)	1898 (1401)	1882 (1352)	0.79
Crystalloids, No. (%)	920 (93.0)	928 (94.0)	0.37
	1838 (1306)	1841 (1298)	0.97
Synthetic colloids, No. (%)	74 (7.5)	56 (5.7)	0.10
	50 (211)	35 (171)	0.09
HES, No. (%)	45 (4.6)	32 (3.2)	0.13
	32 (178)	21 (139)	0.15
Gelatines, No. (%)	31 (3.1)	24 (2.4)	0.34
	17 (104)	13 (102)	0.38
Dextran, No. (%)	1 (0.1)	0 (0.0)	0.99
	1 (15.9)	0 (0)	0.32

PEEP, positive end-expiratory pressure; HES, hydroxyl-ethyl -starch containing solution.

eTable 4. Types of intraoperative vasoactive drugs

	High PEEP n=989	Low PEEP n=987	P value
Need for vasoactive drugs	491 (49.6)	439 (44.5)	0.02
Dopamine, No. (%)	0 (0.0)	2 (0.2)	1.00
(mg)	0.0 (0.0)	0.0 (1.5)	0.16
Norepinephrine, No. (%)	164 (16.6)	154 (15.6)	0.55
(mg)	0.3 (1.2)	0.2 (1.2)	0.36
Phenylephrine, No. (%)	232 (23.5)	190 (19.3)	0.02
(mg)	0.7 (5.5)	0.5 (4.4)	0.26
Epinephrine, No. (%)	14 (1.4)	15 (1.5)	0.85
(mg)	0.0 (0.7)	0.0 (0.7)	0.37
Use of inotropic drug			
Dobutamine, No. (%)	1 (0.1)	1 (0.1)	1.00
(mg)	0.0 (0.9)	0.0 (0.0)	0.32

PEEP, positive end-expiratory pressure.

eTable 5. Types of anesthetic, muscle paralyzing and reversal agents

	High PEEP n=989	Low PEEP n=987	P value
Volatile anesthetics, No.(%)			
Desflurane	469 (47.4)	468 (47.4)	1.00
Sevoflurane	401 (40.5)	411 (41.6)	0.62
Isoflurane	40 (4.0)	37 (3.7)	0.73
Intravenous anesthetics, No.(%)			
Propofol	927 (93.7)	911 (92.3)	0.21
Midazolam	278 (28.1)	287 (29.1)	0.63
Thiopental	37 (3.7)	40 (4.1)	0.72
Ketamine	144 (14.6)	136 (13.8)	0.62
Etomidate	7 (0.7)	7 (0.7)	1.00
Dexmedetomidine	78 (7.9)	79 (8.0)	0.92
Others	38 (3.8)	31 (3.1)	0.40
Analgesics, No.(%)			
Morphine	160 (16.2)	158 (16.0)	0.92
Alfentanil	7 (0.7)	8 (0.8)	0.79
Fentanyl	539 (54.5)	570 (57.8)	0.15
Sufentanil	239 (24.2)	208 (21.1)	0.10
Remifentanyl	331 (33.5)	339 (34.3)	0.68
Lidocaine	320 (32.4)	313 (31.7)	0.76
Procaine	2 (0.2)	0 (0.0)	0.48
NSAIDS	236 (23.9)	239 (24.2)	0.85
Others	308 (31.1)	280 (28.4)	0.18
Muscle paralyzing agents, No.(%)			
Atracurium	27 (2.7)	24 (2.4)	0.68
Cis-atracurium	66 (6.7)	51 (5.2)	0.16
Mivacurium	1 (0.1)	2 (0.2)	1.00
Pancuronium	5 (0.5)	2 (0.2)	0.45
Vecuronium	47 (4.8)	48 (4.9)	0.91
Rocuronium	847 (85.6)	848 (85.9)	0.86
Succinylcholine	240 (24.3)	234 (23.7)	0.77
Others	4 (0.4)	3 (0.3)	0.71
Reversal of muscle paralysis agents, No.(%)			
Sugammadex	438 (44.3)	434 (44.0)	0.89
Cholinesterase inhibitors	290 (29.3)	295 (29.9)	0.78

PEEP, positive end-expiratory pressure.

eTable 6. Further characteristics of surgery

	High PEEP n=989	Low PEEP n=987	P value
Priority of surgery, No.(%)			
Elective	962 (97.8)	958 (97.6)	0.64
Emergency	13 (1.3)	11 (1.1)	
Urgent	9 (0.9)	13 (1.3)	
Positioning during surgery, No.(%)			
Supine	357 (36.2)	367 (37.2)	0.69
Trendelenburg	86 (8.7)	69 (7.0)	
Reverse Trendelenburg	486 (49.3)	491 (49.8)	
Lithotomy	17 (1.7)	16 (1.6)	
Seated	39 (4.0)	43 (4.4)	
Surgical wound classification, No.(%)			
Clean	514 (52.1)	525 (53.2)	0.46
Clean-contaminated	461 (46.8)	450 (45.6)	
Contaminated	10 (1.0)	7 (0.7)	
Dirty	1 (0.1)	4 (0.4)	
Intra-abdominal pressure during laparoscopy - (mmHg), mean (SD)	14.7 (3.6)	14.5 (2.1)	0.17

PEEP, positive end-expiratory pressure.

eTable 7. Assessment of postoperative pain and dyspnea

	High PEEP n=989	Low PEEP n=987	P value
VAS dyspnea – cm, mean (SD), No.			
Day 1	1.8 (1.5), 912	1.8 (1.4), 917	0.53
Day 2	1.4 (1.0), 840	1.5 (1.0), 844	0.57
Day 3	1.4 (0.9), 619	1.3 (0.8), 617	0.68
Day 4	1.3 (0.9), 461	1.3 (0.8), 459	0.79
Day 5	1.2 (0.9), 336	1.2 (0.8), 330	0.78
VAS thoracic – cm, mean (SD), No.			
Day 1	1.6 (1.3), 914	1.7 (1.4), 913	0.52
Day 2	1.3 (0.8), 838	1.4 (3.9), 841	0.34
Day 3	1.2 (0.6), 620	1.2 (0.6), 617	0.94
Day 4	1.1 (0.5), 461	1.2 (0.5), 459	0.20
Day 5	1.1 (0.4), 336	1.1 (0.5), 330	0.16
VAS abdominal rest pain – cm, mean (SD), No.			
Day 1	2.7 (1.8), 950	2.9 (1.9), 958	0.10
Day 2	2.1 (1.5), 881	2.1 (1.4), 889	0.31
Day 3	1.8 (1.3), 647	1.8 (1.2), 650	0.94
Day 4	1.6 (1.2), 469	1.6 (1.0), 475	0.75
Day 5	1.5 (1.0), 346	1.5 (1.0), 337	0.83
VAS abdominal incident pain – cm, mean (SD), No.			
Day 1	3.7 (2.4), 916	3.7 (2.3), 929	0.52
Day 2	2.8 (2.0), 837	2.9 (1.9), 857	0.51
Day 3	2.4 (1.7), 615	2.4 (1.7), 627	0.55
Day 4	2.0 (1.5), 458	2.1 (1.5), 459	0.69
Day 5	1.9 (1.4), 339	1.9 (1.5), 333	0.58

PEEP, positive end-expiratory pressure; VAS, visual analogue scale.

eTable 8. Per-protocol analysis

	High PEEP n=917	Low PEEP n=912	Effect Estimate 95% CI	P value
Primary outcome				
Postoperative pulmonary complications	186 / 917 (20.3)	209 / 912 (22.9)	0.92 (0.82 – 1.04)	0.17
Mild respiratory failure	119 / 917 (13.0)	140 / 912 (15.4)	0.90 (0.78 – 1.04)	0.15
Moderate respiratory failure	38 / 917 (4.1)	51 / 912 (5.6)	0.84 (0.66 – 1.07)	0.15
Severe respiratory failure	25 / 917 (2.7)	35 / 912 (3.8)	0.83 (0.61 – 1.11)	0.18
ARDS	3 / 917 (0.3)	1 / 912 (0.1)	1.50 (0.84 – 2.64)	0.62
Bronchospasm	11 / 917 (1.2)	10 / 912 (1.1)	1.04 (0.69 – 1.57)	0.84
New pulmonary infiltrates	14 / 917 (1.5)	14 / 912 (1.5)	0.99 (0.68 – 1.44)	0.99
Pulmonary infection	10 / 917 (1.1)	9 / 912 (1.0)	1.05 (0.68 – 1.61)	0.83
Aspiration pneumonitis	2 / 917 (0.2)	1 / 912 (0.1)	1.33 (0.59 – 2.96)	1.00
Pleural effusion	38 / 917 (4.1)	18 / 912 (2.0)	1.37 (1.14 – 1.65)	0.01
Atelectasis	38 / 917 (4.1)	50 / 912 (5.5)	0.85 (0.67 – 1.09)	0.18
Cardiopulmonary edema	15 / 917 (1.6)	7 / 912 (0.8)	1.36 (1.02 – 1.82)	0.09
Pneumothorax	1 / 917 (0.1)	2 / 912 (0.2)	0.66 (0.13 – 3.29)	0.62

PEEP, positive end-expiratory pressure; CI, confidence interval; ARDS, acute respiratory distress syndrome.

eTable 9. Sensitivity analysis for the primary outcome

	High PEEP n=989	Low PEEP n=987	Odds ratio 95% CI	P value
Random-effect model	—	—	0.86 (0.68 – 1.08)*	0.19
Count events	0.35 (0.88)	0.38 (0.86)		
Median (IQR)	0 (0 – 0)	0 (0 – 0)	0.88 (0.72 – 1.09)£	0.25
Common effect GEE	—	—	0.93 (0.80 – 1.08)¥	0.37
Treatment-component interaction GEE	—	—	—	0.11**
Average relative effect GEE	—	—	0.99 (0.94 – 1.05)££	0.98

PEEP: positive end expiratory pressure; IQR: interquartile range; GEE: generalized estimating equations

* 95% confidence intervals and p values calculated in a mixed-effect model with study sites as random-effects

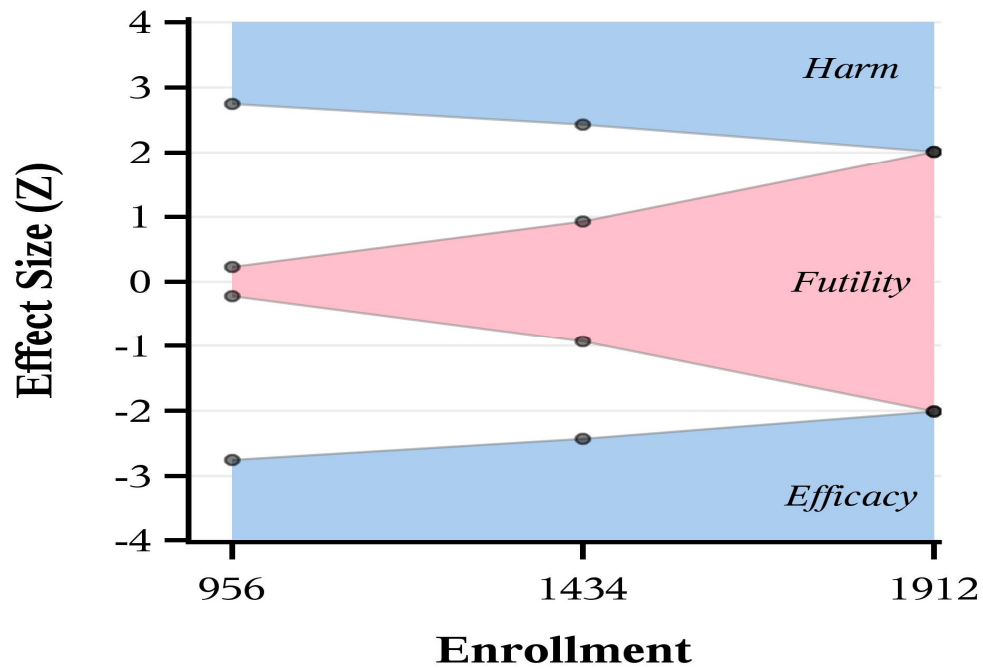
£ 95% confidence intervals calculated with proportional odds logistic regression and p values calculated Wilcoxon rank-sum test

¥ 95% confidence intervals and p values calculated in a common effect GEE model (estimating a single treatment effect across all 12 components)

** p value calculated in a GEE model (test whether the treatment effect differs across the 12 components)

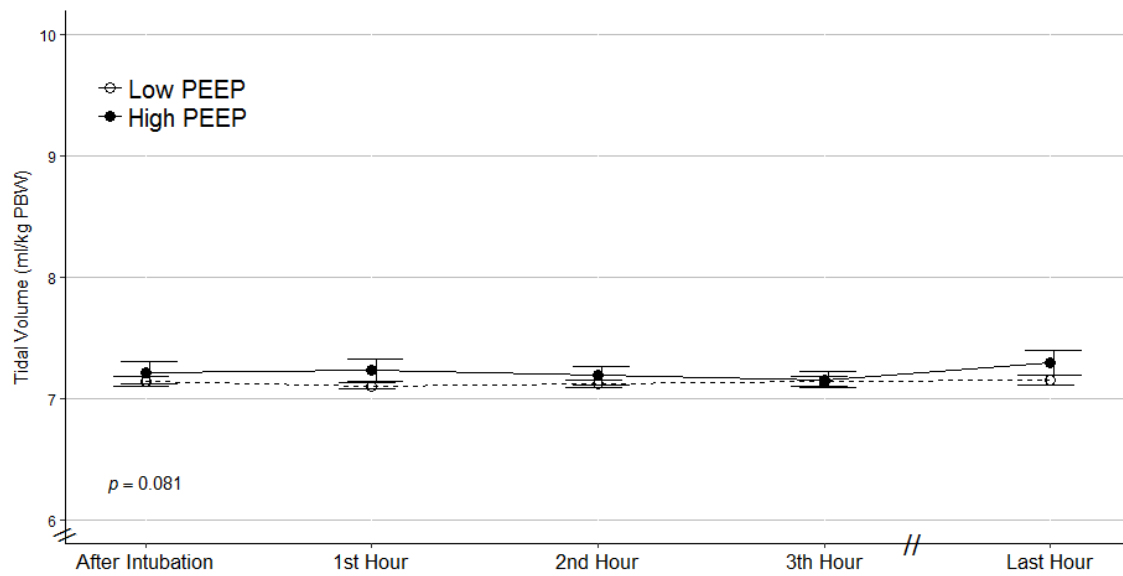
££ p value calculated in a GEE model (estimating, then averaging, the 12 distinct treatment effects)

eFigure 1. Effect size (Z) according to the enrollment of patients in the trial



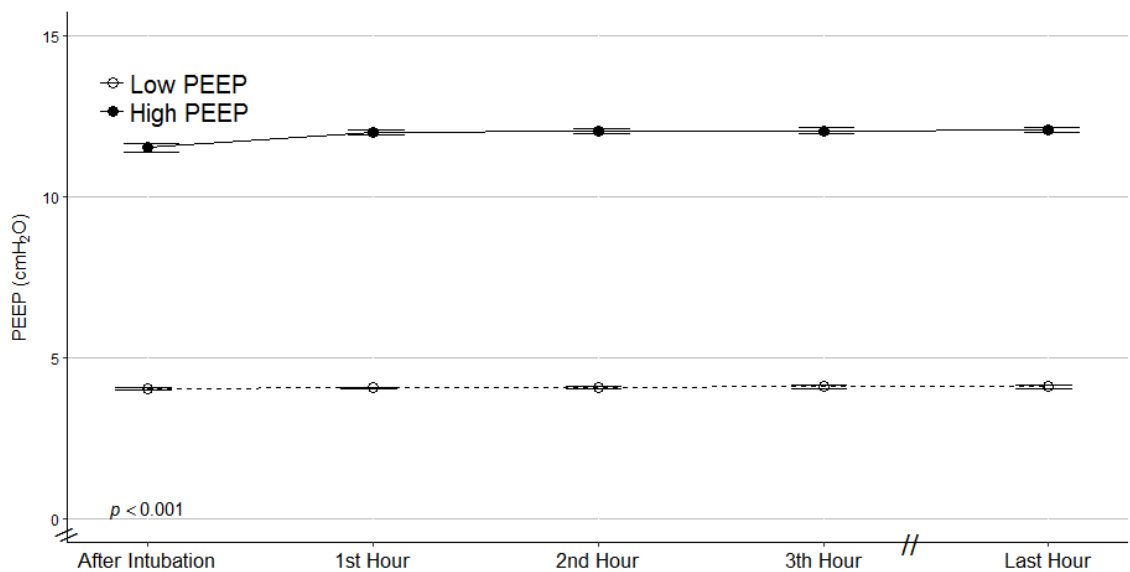
Numbers of patients enrolled in the figure do not include dropouts.

eFigure 2. Tidal volume during anesthesia



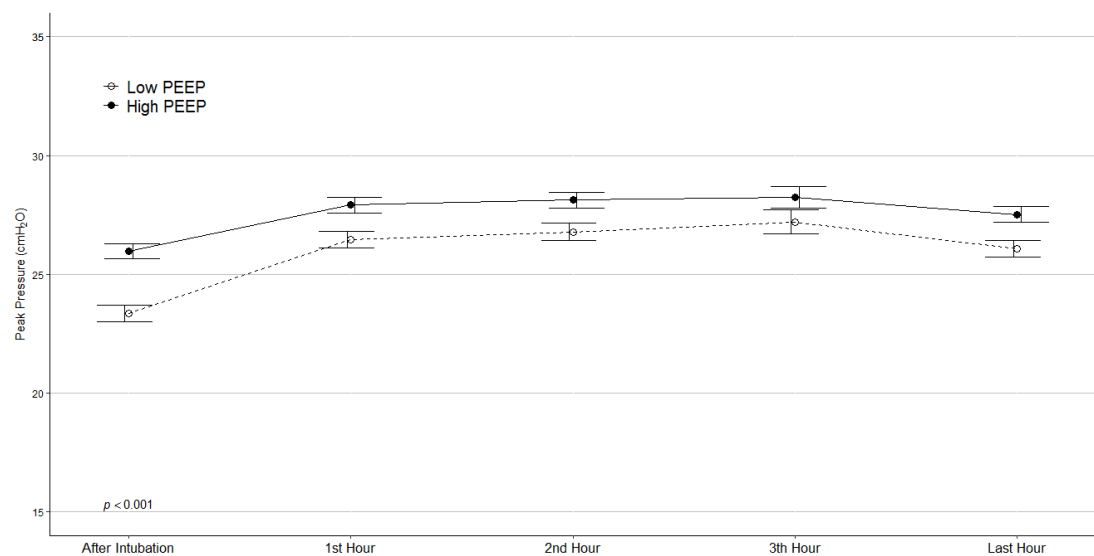
PEEP, positive end-expiratory pressure. Dots are mean and error bars are 95% confidence intervals. P-value from a time vs. group interaction calculated in a mixed-effect linear model.

eFigure 3. Positive end-expiratory pressure during anesthesia



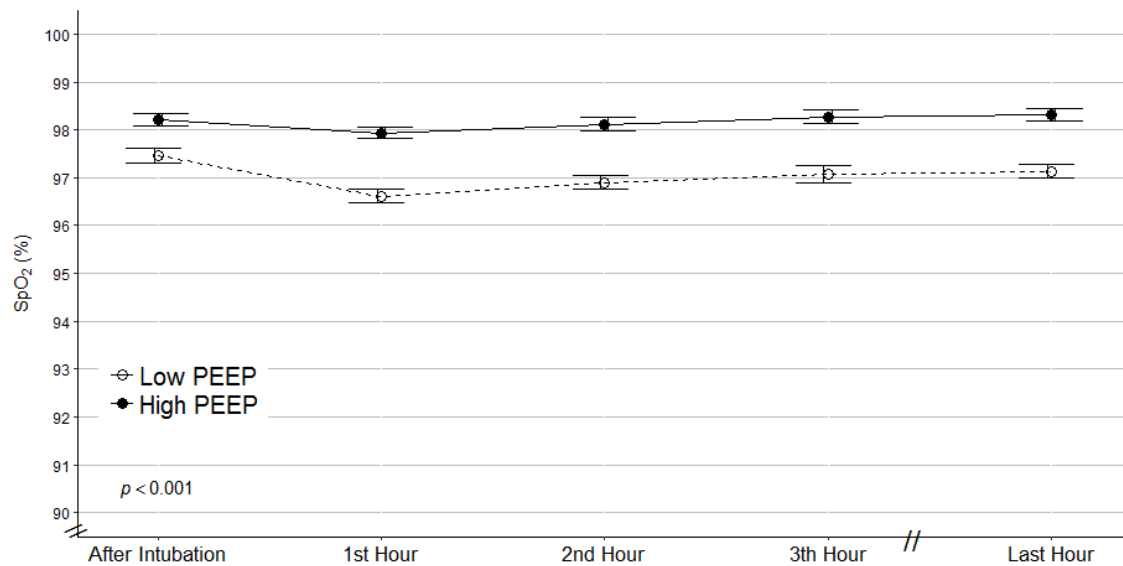
PEEP, positive end-expiratory pressure. Dots are mean and error bars are 95% confidence intervals. P-value from a time vs. group interaction calculated in a mixed-effect linear model.

eFigure 4. Peak pressure during anesthesia



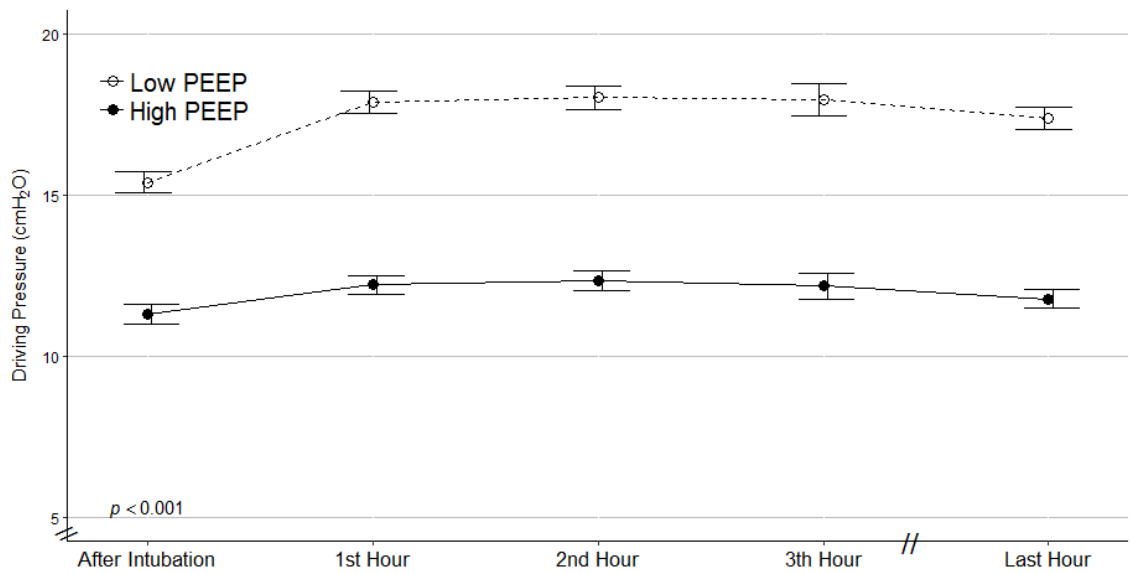
Inspiratory peak pressure values were read from the anesthesia ventilator. PEEP, positive end-expiratory pressure. Dots are mean and error bars are 95% confidence intervals. P-value from a time vs. group interaction calculated in a mixed-effect linear model.

eFigure 5. Peripheral oxyhemoglobin saturation during anesthesia



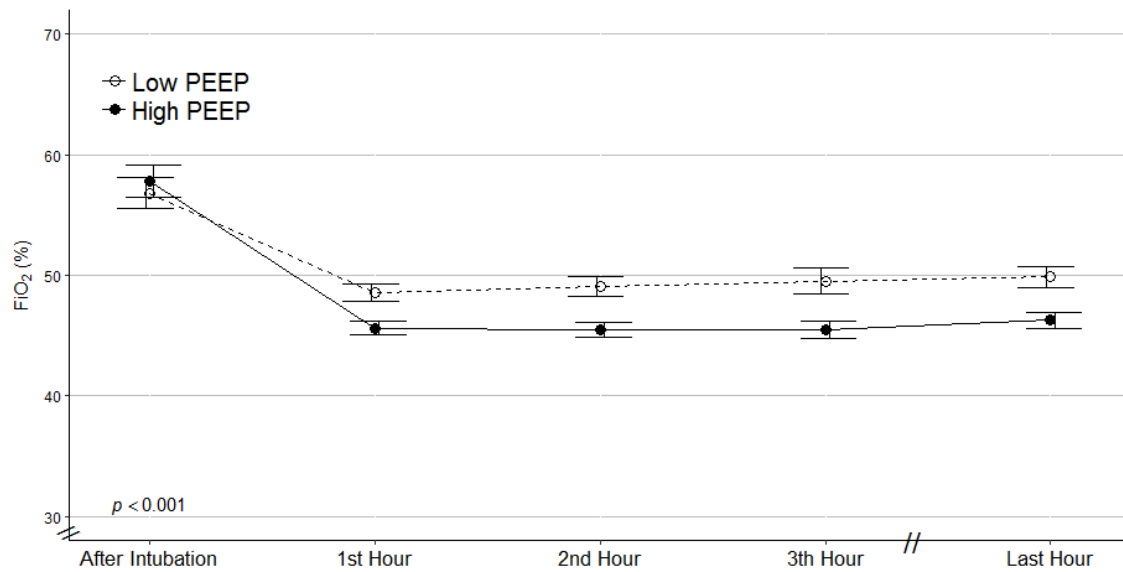
PEEP, positive end-expiratory pressure. Dots are mean and error bars are 95% confidence intervals. P-value from a time vs. group interaction calculated in a mixed-effect linear model.

eFigure 6. Driving pressure during anesthesia



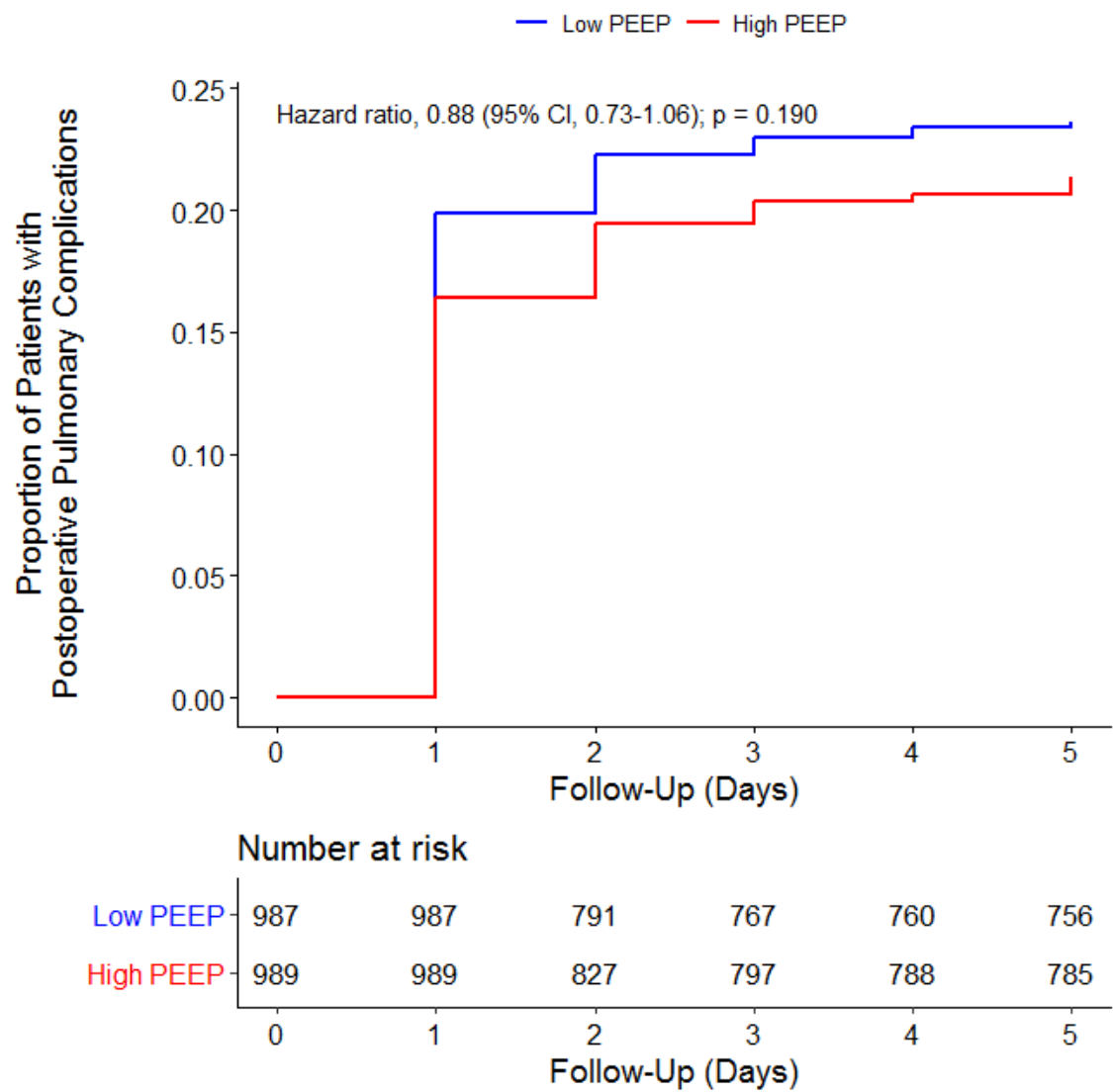
Driving pressure was calculated as airway plateau pressure minus positive end-expiratory pressure. PEEP, positive end-expiratory pressure. Dots are mean and error bars are 95% confidence intervals. P-value from a time vs. group interaction calculated in a mixed-effect linear model.

eFigure 7. Fraction of inspiratory oxygen during anesthesia



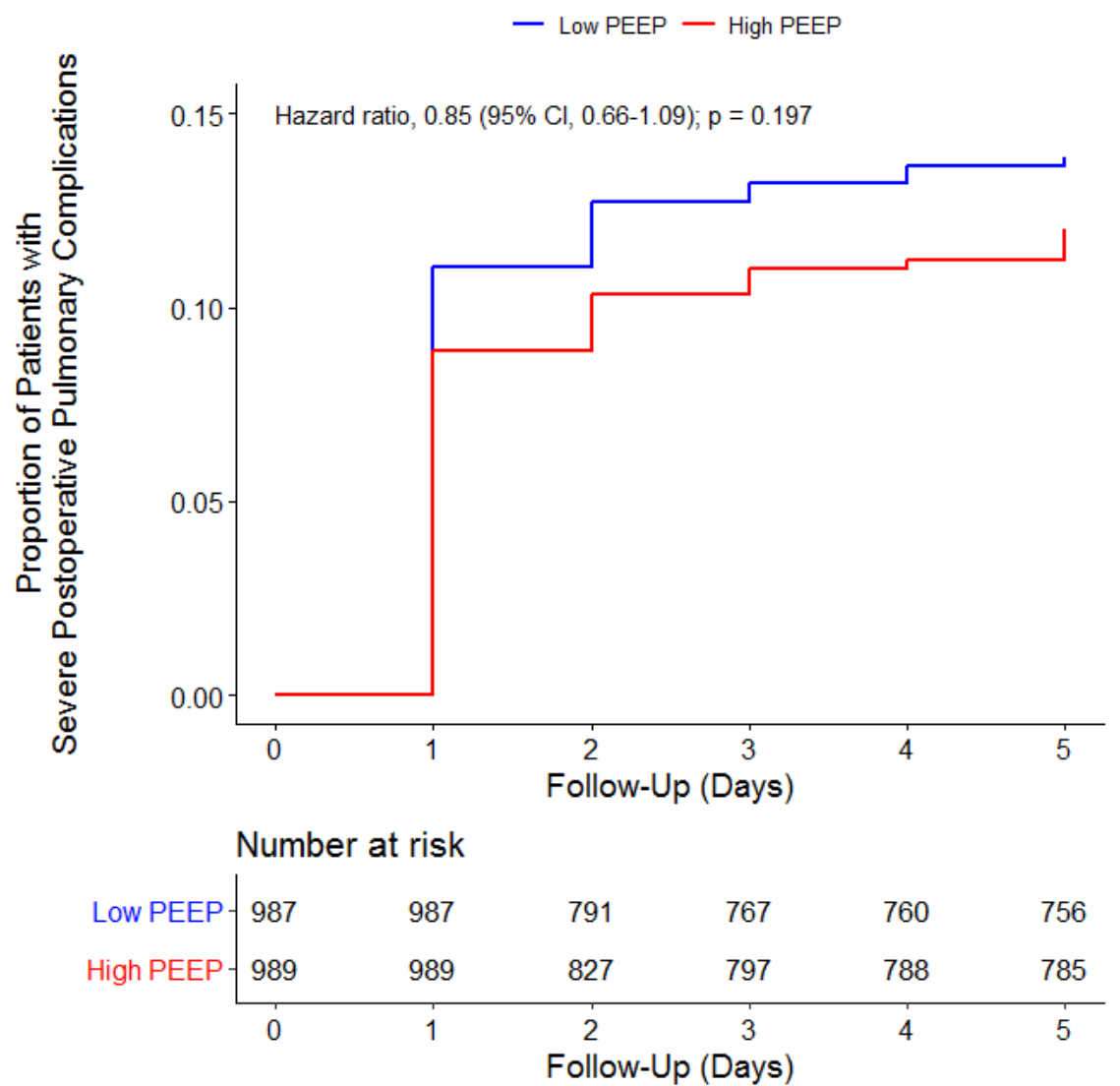
PEEP, positive end-expiratory pressure. Dots are mean and error bars are 95% confidence intervals. P-value from a time vs. group interaction calculated in a mixed-effect linear model.

eFigure 8. Probability of postoperative pulmonary complications in the first 5 postoperative days



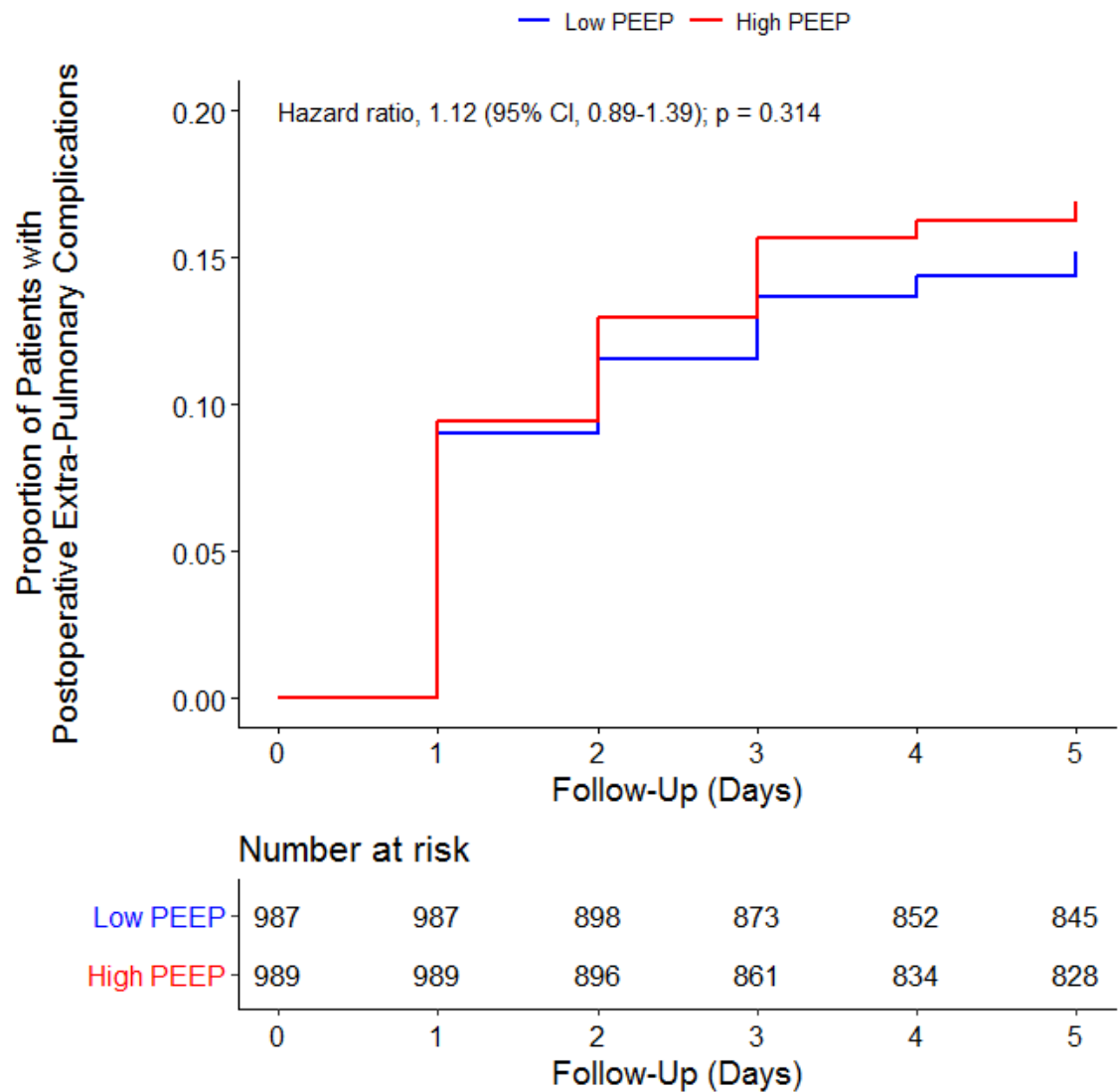
Median observation time was not computed for incidence of postoperative pulmonary complications because the maximum value observed was 0.23. With a median follow-up of 4 (2 – 5) days, the rate of postoperative pulmonary complications within the first 5 postoperative days was 21.3% in the high PEEP group and 23.6% in the low PEEP group (hazard ratio for postoperative pulmonary complications, 0.88; 95% confidence interval 0.73 to 1.06; P=0.190). P value for the Schoenfeld residuals was 0.05.

eFigure 9. Probability of severe postoperative pulmonary complications in the first 5 postoperative days



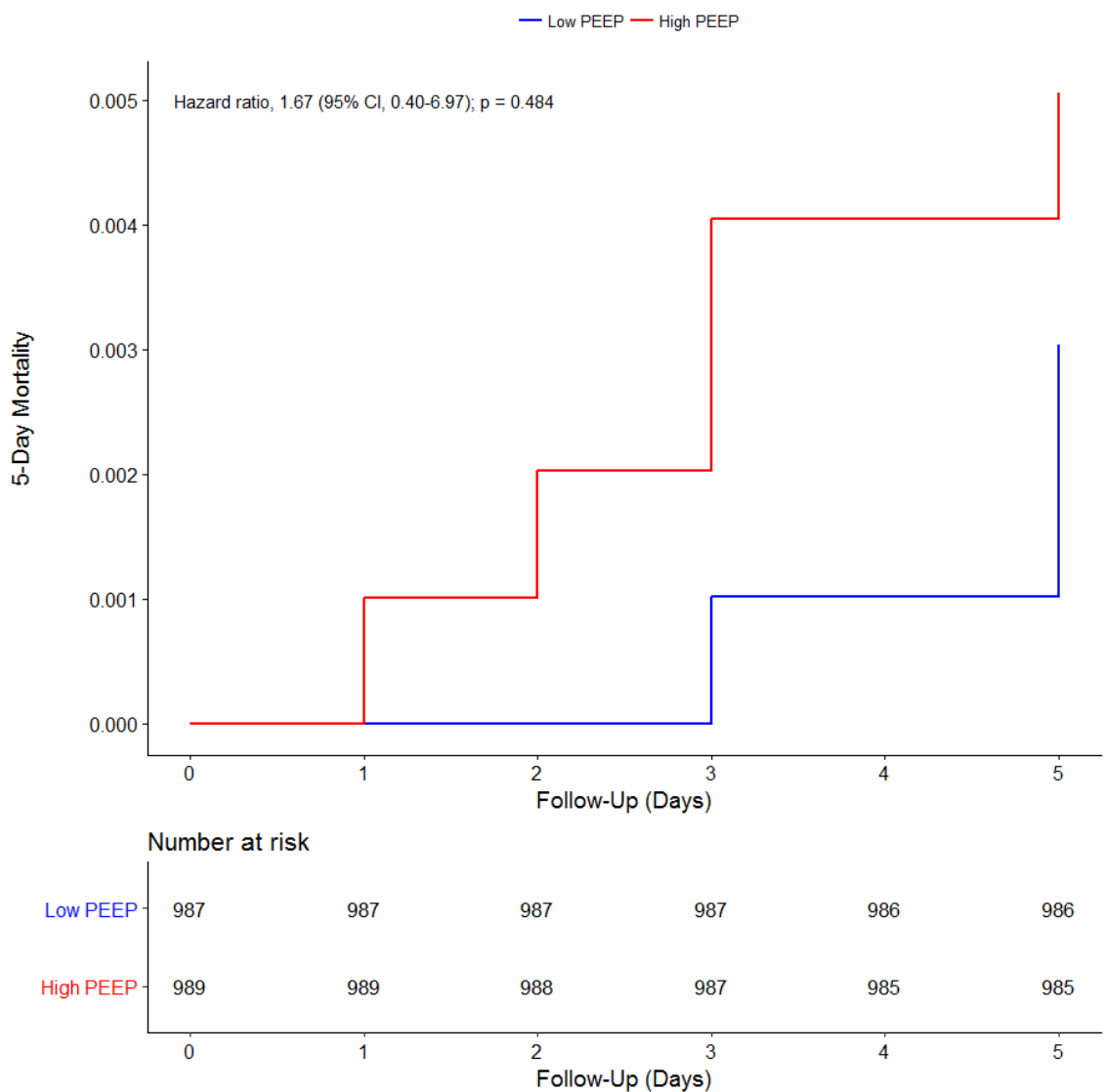
With a median follow-up of 4 (2 – 5) days, the rate of severe postoperative pulmonary complications within the first 5 postoperative days was 11.7% in the high positive end-expiratory pressure (PEEP) group and 13.6% in the low PEEP group (hazard ratio for severe postoperative pulmonary complications, 0.85; 95% confidence interval 0.66 to 1.09; P=0.197). P value for the Schoenfeld residuals was 0.28.

eFigure 10. Probability of postoperative extra-pulmonary complications in the first 5 postoperative days



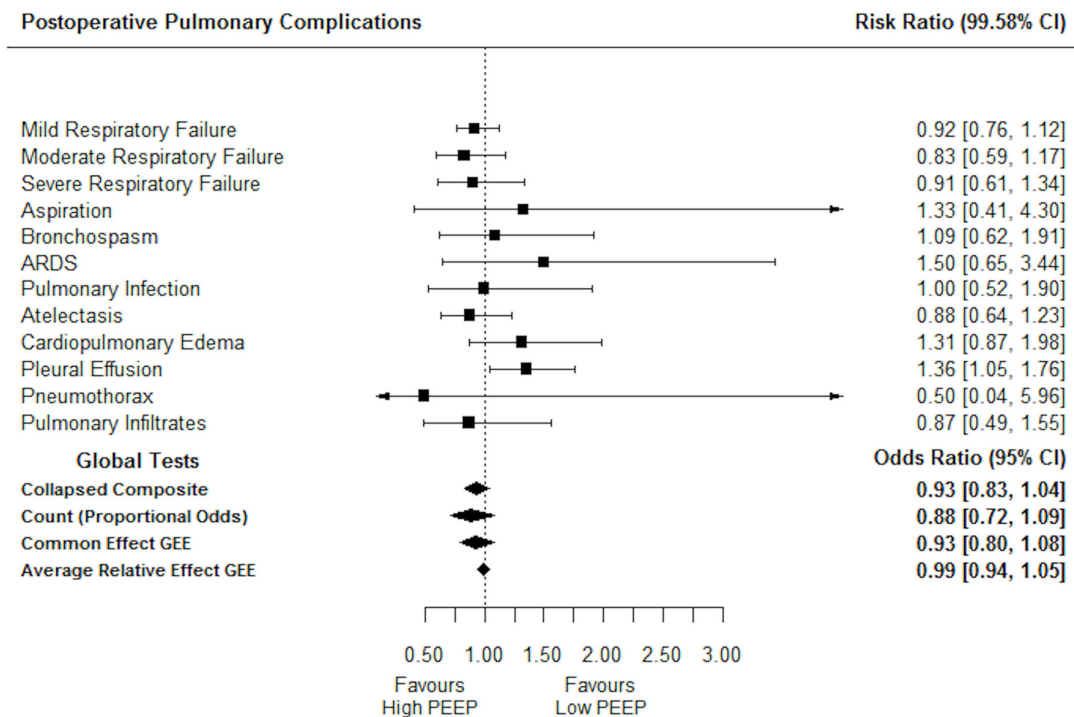
With a median follow-up of 4 (2 – 5) days, the rate of postoperative extra-pulmonary complications within the first 5 postoperative days was 16.9% in the high positive end-expiratory pressure (PEEP) group and 15.2% in the low PEEP group (hazard ratio for postoperative extra-pulmonary complications, 1.12; 95% confidence interval 0.89 to 1.39; P=0.314). P value for the Schoenfeld residuals was 0.67.

eFigure 11. Probability of death in the first 5 postoperative days



With a median follow-up of 4 (2 – 5) days, the rate of postoperative extra-pulmonary complications within the first 5 postoperative days was 0.5% in the high positive end-expiratory pressure (PEEP) group and 0.3% in the low PEEP group (hazard ratio for 5-day mortality, 1.67; 95% confidence interval 0.40 to 6.97; P=0.484). P value for the Schoenfeld residuals was 0.14.

eFigure 12. Results of the sensitivity analysis for the primary endpoint



CI, confidence interval; ARDS, acute respiratory distress syndrome; GEE, generalized estimating equations; PEEP, positive end-expiratory pressure.

eReferences

1. Canet J, Gallart L, Gomar C, et al. Prediction of postoperative pulmonary complications in a population-based surgical cohort. *Anesthesiology* 2010;113:1338-50.
2. Force ADT, Ranieri VM, Rubenfeld GD, et al. Acute respiratory distress syndrome: the Berlin Definition. *JAMA* 2012;307:2526-33.
3. Morrison LJ, Deakin CD, Morley PT, et al. Part 8: Advanced life support: 2010 International Consensus on Cardiopulmonary Resuscitation and Emergency Cardiovascular Care Science With Treatment Recommendations. *Circulation* 2010;122:S345-421.
4. Levy MM, Fink MP, Marshall JC, et al. 2001 SCCM/ESICM/ACCP/ATS/SIS International Sepsis Definitions Conference. *Intensive Care Med* 2003;29:530-8.
5. Thygesen K, Alpert JS, White HD, et al. Universal definition of myocardial infarction. *Circulation* 2007;116:2634-53.
6. Bellomo R, Ronco C, Kellum JA, Mehta RL, Palevsky P, Acute Dialysis Quality Initiative w. Acute renal failure - definition, outcome measures, animal models, fluid therapy and information technology needs: the Second International Consensus Conference of the Acute Dialysis Quality Initiative (ADQI) Group. *Crit Care* 2004;8:R204-12.
7. Levi M, Toh CH, Thachil J, Watson HG. Guidelines for the diagnosis and management of disseminated intravascular coagulation. *British Committee for Standards in Haematology. British journal of haematology* 2009;145:24-33.
8. Reintam A, Parm P, Kitus R, Starkopf J, Kern H. Gastrointestinal failure score in critically ill patients: a prospective observational study. *Crit Care* 2008;12:R90.
9. Du ZG, Wei YG, Chen KF, Li B. An accurate predictor of liver failure and death after hepatectomy: a single institution's experience with 478 consecutive cases. *World journal of gastroenterology* 2014;20:274-81.
10. Wlodzimirow KA, Eslami S, Abu-Hanna A, Nieuwoudt M, Chamuleau RA. Systematic review: acute liver failure - one disease, more than 40 definitions. *Alimentary pharmacology & therapeutics* 2012;35:1245-56.